Simulation Analysis

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2025-04-19

Introduction and Overview

*I need to write this once I figure out what goes here

Data Generating Process

The simulations are a fully factored experiment.

Block Level Fixed Covariates
Number of Blocks: 5; ATE: .2; ICC: .2; Average Number of Observations Per Cluster: 20; Proportion of Clusters Treated: .5 (this is fixed across blocks); Number of District Level Covariates: 0; How much does the average cluster size vary across blocks (coefficient of variation)?: 0

Block Level Varying Covariates (Factors that are varied across simulations)

Three groups of block level parameters Blocks all the same, Blocks all messy, and Blocks all reasonable

Blocks all the same: How much do number clusters per block vary (coefficient of variation)?: 0 (none); ATE SD: 0; Correlation Between Treatment Effect and Cluster Size: 0; Correlation Between Average Number of Observations Per Cluster and Total Number of Clusters: 0; Correlation of Proportion of Clusters Treated per Block and Impact: 0; Correlation Between Outcome Intercept and Cluster Size: 0

Blocks all messy: How much do number clusters per block vary (coefficient of variation)?: .85 (big); ATE SD: .6; Correlation Between Treatment Effect and Cluster Size: .8; Correlation Between Average Number of Observations Per Cluster and Total Number of Clusters: .8; Correlation of Proportion of Clusters Treated per Block and Impact: .8; Correlation Between Outcome Intercept and Cluster Size: .8

Blocks all reasonable: How much do number clusters per block vary (coefficient of variation)?: .33 (average); ATE SD: .1; Correlation Between Treatment Effect and Cluster Size: .2; Correlation Between Average Number of Observations Per Cluster and Total Number of Clusters: .1; Correlation of Proportion of Clusters Treated per Block and Impact: 0; Correlation Between Outcome Intercept and Cluster Size: .1

Each of the rest of these factors don't vary as a group but vary each individually and are fully factored with all other factors:

Average Number of Clusters per Block: 7,10,16

Cluster Level Varying Parameters

Correlation of Impact and Size of Cluster: 0, .8

Cluster Level Treatment Effect Standard Deviation: 0, .3

Cluster Level ICC: 0, .2

Number of Observations Per Cluster Varies: True, False

How much does the size of clusters vary (coefficient of variation)?: 0, .33, .6

Number of School-Level Covariates: 0, 1, 5

[School Level Covariate R Squared is always .5]

What does the bias look like for each estimator

*The next thing to do is to move all the simulation questions I'm answering to my emails and also to attach the page numbers to the analysis

Main Bullet Points:

*The estimators theory says are unbiased are indeed consistently unbiased (or have very low bias across the full range of simulations)

*The LRapw−FIpw−db has bias as large as -.025, which seems from the statistical model to be driven by the interactions of a large treatment effect standard deviation across clusters, high variability in cluster size, and correlations between treatment effect and cluster size. This is consitent with theory.

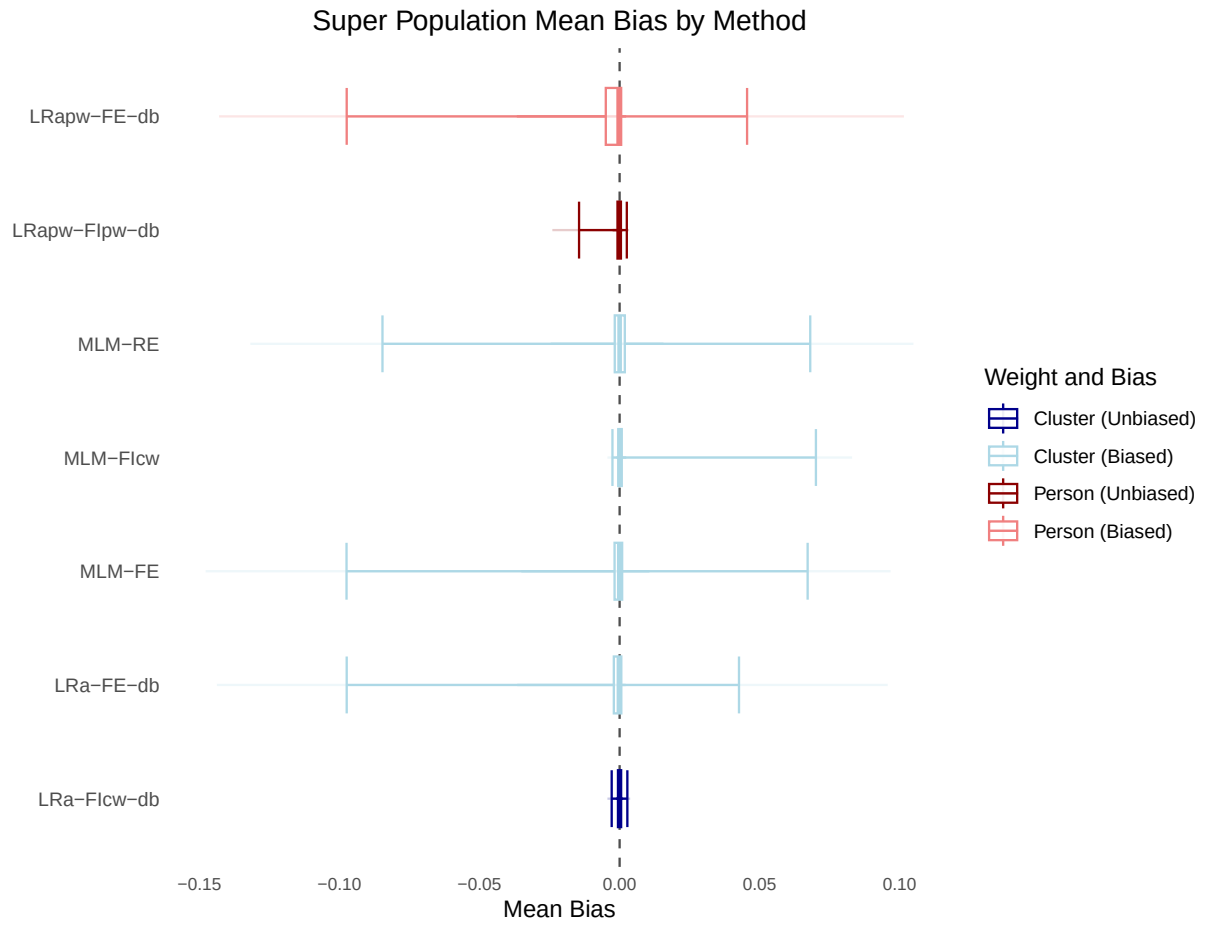
*For the various estimators that are biased, the mean bias is mostly driven by the different blocks being different from each other

Description of Analysis

*The graph is a boxplot of mean bias across the different scenarios by method. The faint horizontal lines go out to the full range, the bars are the 1% to 99%, and the boxes are the 10% to 90%.

*The first table contains the regression coefficients from a regression that predicts the magnitude of the mean bias using different data generating parameters. Each regression is run separately by estimator and mean bias is the mean bias against the targeted estimator.

*The second table contains the DGPs that produce that largest magnitude of mean bias for each method, conditional on the magnitude of the mean bias being at least .1 . Where again mean bias is the mean bias against the targeted estimator.



The Effect of Different Data Generating Parameters on the Magnitude of Mean Bias

Term	LRa-FE-db	LRa-FIcw-db	LRapw-FE-db	LRapw-FIpw-db	MLM-FE	MLM-FIcw	MLM-RE
(Intercept)	.007.	<.001*	.009*	.002***	.009*	.005**	.009*
block_typeS	<.001	<.001	<.001	<.001	<.001	<.001	.001
block_typeL	.036***	<.001	.037***	<.001	.042***	.001	.036***
J.barS	-.003	<.001	-.003	<.001	-.004	-.001	-.005
J.barL	-.002	<.001	-.002	.001**	-.004	-.002	-.003
size.impact.cor.schoolL	-.005	<.001	-.006*	-.003***	-.009**	-.008***	-.008**
sd.schoolL	<.001	<.001	-.002	-.002***	-.005	-.008***	-.007*
ICC.schoolL	-.007*	.001***	-.008*	.001**	<.001	.010***	<.001
variable.nTRUE	-.006.	<.001	-.007*	-.002***	-.010**	-.008***	-.009**
cv.schoolS	-.006	<.001	-.006.	-.001***	-.008.	-.004*	-.008*
cv.schoolL	-.001	<.001	-.002	-.002***	-.005	-.009***	-.007.
num.cov.schoolS	-.005	<.001	-.005	<.001	-.005	<.001	-.004
num.cov.schoolL	.001	<.001	.001	<.001	<.001	<.001	.001
block_typeS:J.barS	<.001	<.001	<.001	<.001	<.001	<.001	-.002
block_typeL:J.barS	<.001	<.001	<.001	<.001	-.001	<.001	.001
block_typeS:J.barL	<.001	<.001	<.001	<.001	<.001	<.001	-.003
block_typeL:J.barL	-.001	<.001	<.001	<.001	-.001	<.001	.004
block_typeS: size.impact.cor.schoolL	<.001	<.001.	<.001	<.001	<.001	<.001	<.001
block_typeL: size.impact.cor.schoolL	-.006**	<.001	-.007**	<.001	-.011***	<.001	-.009***
block_typeS:sd.schoolL	<.001	<.001	<.001	<.001	<.001	<.001	<.001
block_typeL:sd.schoolL	-.002	<.001	-.002	<.001	-.007**	<.001	-.007**

Term	LRa-FE-db	LRa-FIcw-db	LRapw-FE-db	LRapw-FIpw-db	MLM-FE	MLM-FIcw	MLM-RE
block_typeS:ICC.schoolL	<.001	<.001	<.001	<.001	<.001	<.001	.002
block_typeL:ICC.schoolL	-.005*	<.001	-.004*	<.001	-.003	<.001	-.005*
block_typeS: variable.nTRUE	<.001	<.001	<.001	<.001	<.001	<.001	<.001
block_typeL: variable.nTRUE	.003	<.001	.001	<.001	-.002	<.001	-.002
block_typeS:cv.schoolS	<.001	<.001	<.001	<.001	<.001	<.001	<.001
block_typeL:cv.schoolS	.003	<.001	.002	<.001	.001	<.001	<.001
block_typeS:cv.schoolL	<.001	<.001	<.001	<.001	<.001	<.001	<.001
block_typeL:cv.schoolL	.003	<.001	.001	<.001	-.003	<.001	-.004
block_typeS: num.cov.schoolS	<.001	<.001	<.001	<.001	<.001	<.001	<.001
block_typeL: num.cov.schoolS	-.003	<.001	-.003	<.001	-.003	<.001	-.002
block_typeS: num.cov.schoolL	<.001	<.001	<.001	<.001	<.001	<.001	.001
block_typeL: num.cov.schoolL	-.001	<.001	-.002	<.001	-.001	<.001	.001
J.barS: size.impact.cor.schoolL	.002	<.001	.002	<.001	.003	.001	.002
J.barL: size.impact.cor.schoolL	.003	<.001	.002	-.001***	.005.	.002	.004
J.barS:sd.schoolL	<.001	<.001	-.001	<.001	.001	.001	.002
J.barL:sd.schoolL	.001	<.001	<.001	-.001***	.003	.001	.003
J.barS:ICC.schoolL	.003	<.001	.003	<.001	.003	<.001	.002
J.barL:ICC.schoolL	.002	<.001*	.002	<.001	<.001	-.001	<.001
J.barS:variable.nTRUE	.003	<.001	.002	<.001	.004	.001	.003
J.barL:variable.nTRUE	.003	<.001	.002	-.001***	.005.	.001	.004.
J.barS:cv.schoolS	.002	<.001	.002	<.001	.002	.001	.003
J.barL:cv.schoolS	-.001	<.001	-.001	<.001	<.001	.001	.001
J.barS:cv.schoolL	-.002	<.001	-.002	<.001	-.001	.001	.001
J.barL:cv.schoolL	-.003	<.001.	-.004	-.001***	-.001	.001	<.001
J.barS:num.cov.schoolS	<.001	<.001	<.001	<.001	<.001	<.001	<.001
J.barL:num.cov.schoolS	<.001	<.001	<.001	<.001	<.001	<.001	-.001
J.barS:num.cov.schoolL	-.003	<.001	-.003	<.001	-.003	<.001	-.002
J.barL:num.cov.schoolL	-.003	<.001	-.002	<.001	-.004	<.001	-.005.
size.impact.cor.schoolL: sd.schoolL	.003	<.001	.005**	.003***	.010***	.011***	.010***
size.impact.cor.schoolL: ICC.schoolL	.002	<.001	.002	<.001	-.002	-.006***	-.002
size.impact.cor.schoolL: variable.nTRUE	<.001	<.001*	.002	.003***	.007***	.011***	.007***
size.impact.cor.schoolL: cv.schoolS	.002	<.001	.003	.001***	.005*	.005***	.005*
size.impact.cor.schoolL: cv.schoolL	.001	<.001	.003	.002***	.008**	.011***	.007**
size.impact.cor.schoolL: num.cov.schoolS	.001	<.001	.001	<.001	.001	<.001	.001
size.impact.cor.schoolL: num.cov.schoolL	-.001	<.001	-.001	<.001	<.001	<.001	-.001
sd.schoolL:ICC.schoolL	<.001	<.001	<.001	<.001	-.005*	-.006***	-.005**
sd.schoolL:variable.nTRUE	.001	<.001	.003	.003***	.008***	.011***	.008***
sd.schoolL:cv.schoolS	-.003	<.001	-.002	.001***	<.001	.005***	.002
sd.schoolL:cv.schoolL	-.002	<.001	<.001	.002***	.006*	.011***	.008***
sd.schoolL: num.cov.schoolS	.001	<.001	.001	<.001	.001	<.001	.002
sd.schoolL: num.cov.schoolL	-.002	<.001	-.002	<.001	-.001	<.001	<.001
ICC.schoolL: variable.nTRUE	.004*	<.001	.004*	<.001	-.001	-.006***	-.001
ICC.schoolL:cv.schoolS	.005*	<.001	.005*	<.001	.003	-.003**	.002
ICC.schoolL:cv.schoolL	.002	<.001	.002	<.001	-.003	-.006***	-.003
ICC.schoolL: num.cov.schoolS	.002	<.001	.003	<.001	.002	<.001	.002
ICC.schoolL: num.cov.schoolL	.001	<.001	.001	<.001	<.001	<.001	.001

Term	LRa-FE-db	LRa-FIcw-db	LRapw-FE-db	LRapw-FIpw-db	MLM-FE	MLM-FIcw	MLM-RE
variable.nTRUE:cv.schoolS	.001	<.001	.002	.001***	.004.	.004***	.005*
variable.nTRUE:cv.schoolL	.002	<.001.	.003	.003***	.009***	.011***	.010***
variable.nTRUE: num.cov.schoolS	<.001	<.001	.001	<.001	.001	<.001	<.001
variable.nTRUE: num.cov.schoolL	.002	<.001	.001	<.001	.002	<.001	<.001
cv.schoolS: num.cov.schoolS	.005.	<.001	.005.	<.001	.005.	<.001	.005.
cv.schoolL: num.cov.schoolS	.002	<.001	.002	<.001	.002	<.001	.002
cv.schoolS: num.cov.schoolL	.002	<.001	.002	<.001	.002	<.001	.001
cv.schoolL: num.cov.schoolL	.002	<.001	.001	<.001	.003	<.001	.002

*Significance: '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1*

Top Five Scenarios with Most Bias by Method

method	bias.s estimand	J.bar	sd school	ICC school	cv school	num.cov school	size.impact cor.school	block type
LRa-FE-db	-0.144	0	0	0	L	L	0	L
LRa-FE-db	-0.143	S	0	L	S	S	0	L
LRa-FE-db	-0.14	0	L	0	S	L	L	L
LRa-FE-db	-0.126	0	0	0	L	L	0	L
LRa-FE-db	-0.123	0	L	L	L	L	0	L
LRapw-FE-db	-0.143	S	0	L	S	S	0	L
LRapw-FE-db	-0.14	0	L	0	S	L	L	L
LRapw-FE-db	-0.134	0	0	0	L	L	0	L
LRapw-FE-db	-0.126	0	0	0	L	L	0	L
LRapw-FE-db	-0.114	0	L	0	0	0	0	L
MLM-FE	-0.148	0	0	0	L	L	0	L
MLM-FE	-0.143	S	0	L	S	S	0	L
MLM-FE	-0.14	0	L	0	S	L	L	L
MLM-FE	-0.126	0	0	0	L	L	0	L
MLM-FE	-0.122	0	L	L	L	L	0	L

method	bias.s estimand	J.bar	sd school	ICC school	cv school	num.cov school	size.impact cor.school	block type
MLM-RE	-0.132	0	0	0	L	L	0	L
MLM-RE	-0.127	S	0	L	S	S	0	L
MLM-RE	-0.117	0	L	0	S	L	L	L
MLM-RE	-0.113	0	0	0	L	L	0	L
MLM-RE	-0.11	S	L	0	L	0	0	L

What does the error look like for each estimator

Main Bullet Points:

*For the person weighted estimators, the unbiased estimator both has less error and is more stable than the biased estimator.

*For the cluster weighted estimators, the unbiased estimator generally has less error than the biased estimators but more instablity.

*The MLM estimators are the most stable, which fits with theory.

*High RMSE and estimate instability is driven by a low number of clusters, the interaction of a low number of clusters and high cluster level ICC, and high cluster level treatment effect variance.

*Re the question on pg. 39, looking at the second boxplot shows that true S.E. do vary across estimators with the true standard error in a given DGP often 40% bigger for some estimates than the median estimate in that DPG.

Description of Analysis:

*The first graph is a boxplot of the normalized RMSE across the different scenarios by method. The normalized RMSE is the RMSE for given method divided by the median RMSE across methods within a given simulation scenario. Therefore each data point is adjusted for the underlying base RMSE from a given set of DGPs and represents the RMSE driven by the estimation method.

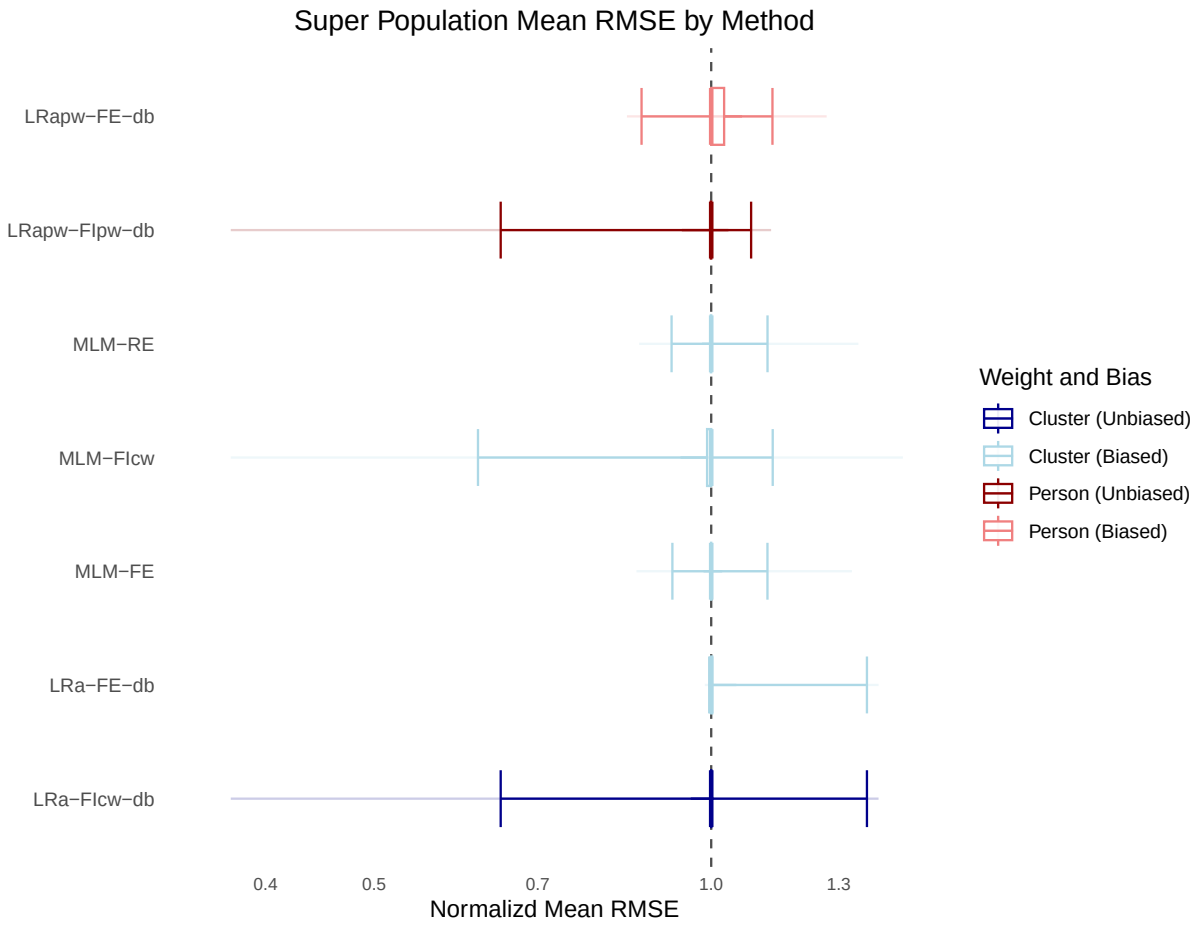
*The second graph is a boxplot of the normalized estimate standard deviation across the different scenarios by method. This is also normalized by dividing by the median estimate standard deviation within a given simulation scenario.

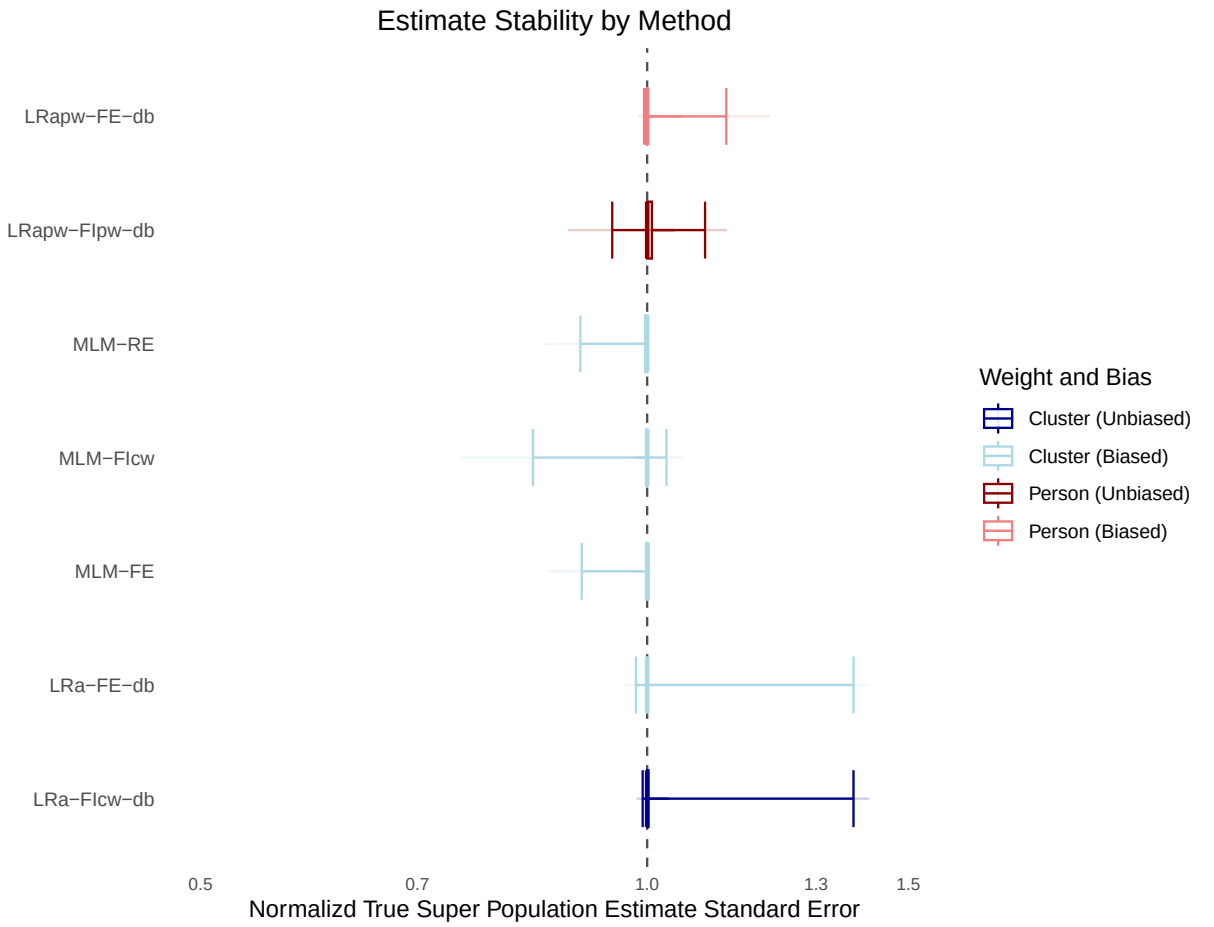
*The first table contains the regression coefficients from a regression that predicts the RMSE using different data generating parameters and the second table is from regressions predicting the estimate standard deviation. Each regression is run separately by estimator and the RMSE is against the targeted estimator.

*The second table contains the DGPs that produce the normalized RMSE (against the targeted estimator) for each method, conditional on the magnitude of the being at least 1.10.

*The third table contains the regression coefficients from a regression that predicts the true superpopulation estimate standard error using different data generating parameters.

*The fourth table contains the DGPs that produce the normalized super population estimator standard error for each method, conditional on the magnitude of the being at least 1.10.


Super Population Mean RMSE by Method
LRapw−FE−db
LRapw−FIpw−db
MLM−RE
MLM−FIcw
MLM−FE
LRa−FE−db
LRa−FIcw−db
0.4
0.5
0.7
1.0
1.3
Normalizd Mean RMSE
Weight and Bias
Cluster (Unbiased)
Cluster (Biased)
Person (Unbiased)
Person (Biased)



The Effect of Different Data Generating Parameters on the RMSE

Term	LRa-FE-db	LRa-FIcw-db	LRapw-FE-db	LRapw-FIpw-db	MLM-FE	MLM-FIcw	MLM-RE
(Intercept)	.071***	.067***	.074***	.070***	.072***	.069***	.072***
block_typeS	.002	.002	.002	.002	.002	.002	.002
block_typeL	.015***	<.001	.014***	<.001	.017***	.001	.015***
J.barS	-.013***	-.011***	-.012***	-.010**	-.013***	-.011***	-.013***
J.barL	-.024***	-.024***	-.022***	-.022***	-.025***	-.025***	-.024***
size.impact.cor.schoolL	-.005.	-.003	-.005	-.003	-.006*	-.005.	-.006.
sd.schoolL	.024***	.026***	.025***	.026***	.024***	.024***	.023***
ICC.schoolL	.090***	.094***	.087***	.090***	.092***	.097***	.093***
variable.nTRUE	.003	.005.	-.005	-.003	-.002	-.001	-.002
cv.schoolS	-.002	.001	-.004	-.001	-.003	<.001	-.003
cv.schoolL	.003	.004	-.007.	-.005.	-.003	-.004	-.003
num.cov.schoolS	-.004	-.001	-.004	-.001	-.004	-.001	-.003
num.cov.schoolL	.001	<.001	.001	<.001	.001	<.001	.001
block_typeS:J.barS	-.002	-.002	-.002	-.002	-.002	-.002	-.002
block_typeL:J.barS	.003	.004.	.002	.003	.002	.003	.003
block_typeS:J.barL	-.001	-.001	-.001	-.001	-.001	-.001	-.001
block_typeL:J.barL	.003	.003	.001	.001	.002	.002	.003
block_typeS: size.impact.cor.schoolL	<.001	<.001	<.001	<.001	<.001	<.001	<.001
block_typeL: size.impact.cor.schoolL	<.001	.002	<.001	.003	-.002	.002	-.001
block_typeS:sd.schoolL	.001	.001	.001	.001	.001	.001	.001
block_typeL:sd.schoolL	<.001	.002	-.001	.002	-.002	.002	-.002
block_typeS:ICC.schoolL	<.001	<.001	<.001	<.001	<.001	<.001	<.001
block_typeL:ICC.schoolL	.001	.009***	<.001	.008***	.001	.009***	.001
block_typeS: variable.nTRUE	.001	.001	.001	.001	.001	.001	.001

Term	LRa-FE-db	LRa-FIcw-db	LRapw-FE-db	LRapw-FIpw-db	MLM-FE	MLM-FIcw	MLM-RE
block_typeL: variable.nTRUE	.002	.002	.008***	.005**	.002	.002	.002
block_typeS:cv.schoolS	-.001	-.001	<.001	<.001	<.001	<.001	<.001
block_typeL:cv.schoolS	.002	.002	.004	.004.	.002	.002	.002
block_typeS:cv.schoolL	-.001	-.001	-.001	-.001	-.001	-.001	-.001
block_typeL:cv.schoolL	.004	.004.	.011***	.008***	.005.	.004.	.004.
block_typeS: num.cov.schoolS	.001	.001	.001	.001	.001	.001	.001
block_typeL: num.cov.schoolS	-.003	-.001	-.003	-.001	-.003	-.001	-.003
block_typeS: num.cov.schoolL	.001	.001	.001	.001	.001	.001	.001
block_typeL: num.cov.schoolL	-.002	-.002	-.002	-.002	-.002	-.002	-.002
J.barS: size.impact.cor.schoolL	.001	<.001	.001	<.001	.001	<.001	.001
J.barL: size.impact.cor.schoolL	-.002	-.003	-.002	-.003	-.001	-.002	-.001
J.barS:sd.schoolL	-.002	-.002	-.002	-.002	-.001	-.002	-.001
J.barL:sd.schoolL	-.003	-.003.	-.003	-.003.	-.002	-.003	-.002
J.barS:ICC.schoolL	-.013***	-.015***	-.014***	-.016***	-.014***	-.016***	-.014***
J.barL:ICC.schoolL	-.028***	-.028***	-.029***	-.030***	-.029***	-.030***	-.030***
J.barS:variable.nTRUE	.001	<.001	.001	<.001	.002	.001	.001
J.barL:variable.nTRUE	-.003	-.004.	-.004.	-.004*	-.001	-.002	-.001
J.barS:cv.schoolS	.001	<.001	.001	<.001	.001	<.001	.001
J.barL:cv.schoolS	<.001	<.001	<.001	<.001	<.001	.001	<.001
J.barS:cv.schoolL	-.003	-.003	-.004	-.003	-.002	-.002	-.002
J.barL:cv.schoolL	-.004.	-.003	-.005.	-.003	-.002	-.001	-.001
J.barS:num.cov.schoolS	<.001	<.001	<.001	<.001	<.001	<.001	<.001
J.barL:num.cov.schoolS	.002	.002	.002	.003	.002	.002	.002
J.barS:num.cov.schoolL	-.002	-.001	-.002	-.001	-.002	-.001	-.002
J.barL:num.cov.schoolL	-.003	-.001	-.003	-.001	-.004	-.002	-.004
size.impact.cor.schoolL: sd.schoolL	<.001	-.001	<.001	-.001	.002	.002	.002
size.impact.cor.schoolL: ICC.schoolL	.003*	.002	.004*	.003.	.002	<.001	.002
size.impact.cor.schoolL: variable.nTRUE	.003	.003.	.003	.003.	.004*	.006***	.005**
size.impact.cor.schoolL: cv.schoolS	.002	<.001	.002	<.001	.002	.001	.002
size.impact.cor.schoolL: cv.schoolL	.002	.001	.001	.001	.004.	.004*	.003.
size.impact.cor.schoolL: num.cov.schoolS	.002	.002	.002	.002	.002	.002	.002
size.impact.cor.schoolL: num.cov.schoolL	.001	.002	.001	.002	.001	.002	.001
sd.schoolL:ICC.schoolL	-.009***	-.010***	-.010***	-.011***	-.012***	-.014***	-.012***
sd.schoolL:variable.nTRUE	-.002	-.003.	-.001	-.001	.001	.001	.001
sd.schoolL:cv.schoolS	-.004*	-.004*	-.004.	-.004.	-.004.	-.003	-.003
sd.schoolL:cv.schoolL	-.004.	-.003	-.001	-.001	<.001	.002	.001
sd.schoolL: num.cov.schoolS	<.001	<.001	<.001	-.001	<.001	-.001	<.001
sd.schoolL: num.cov.schoolL	-.002	-.001	-.002	-.001	-.001	-.001	-.001
ICC.schoolL: variable.nTRUE	-.001	-.003.	.009***	.008***	.002	<.001	.001
ICC.schoolL:cv.schoolS	.001	-.001	.004.	.002	.001	<.001	.001
ICC.schoolL:cv.schoolL	-.005*	-.006**	.008***	.007***	-.001	-.002	-.001
ICC.schoolL: num.cov.schoolS	-.001	-.002	-.001	-.002	-.001	-.002	-.001
ICC.schoolL: num.cov.schoolL	-.001	-.001	-.001	-.002	-.001	-.001	-.001
variable.nTRUE:cv.schoolS	.002	.001	.004*	.003.	.001	.001	.001
variable.nTRUE:cv.schoolL	.015***	.015***	.016***	.014***	.008***	.008***	.008***
variable.nTRUE: num.cov.schoolS	-.004.	-.004*	-.004.	-.004*	-.004.	-.004*	-.004.
variable.nTRUE: num.cov.schoolL	-.001	-.001	-.001	-.001	-.001	-.001	-.001
cv.schoolS: num.cov.schoolS	.007**	.005*	.007**	.005*	.007**	.005*	.006*
cv.schoolL: num.cov.schoolS	.006*	.005*	.006*	.005*	.006*	.005*	.006*
cv.schoolS: num.cov.schoolL	.002	.001	.003	.001	.002	.001	.002
cv.schoolL: num.cov.schoolL	.003	.002	.003	.002	.004	.002	.003

*Significance: '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1*

Top Five Scenarios with Largest Normalized RMSE by Method

method	RMSE.s.estimand.norm	RMSE.sestimand	J.bar	sd school	ICC school	cv school	num.cov school	size.impact cor.school	block type
LRa-FE-db	1.41	0.0635	L	0	0	L	S	0	0
LRa-FE-db	1.4	0.0632	L	0	0	L	0	L	0
LRa-FE-db	1.4	0.0635	L	0	0	L	S	L	0
LRa-FE-db	1.4	0.0633	L	0	0	L	L	L	0
LRa-FE-db	1.4	0.0811	S	0	0	L	0	L	0
LRapw-FE-db	1.27	0.098	L	0	0	L	S	0	L
LRapw-FE-db	1.25	0.113	L	L	0	S	S	L	L
LRapw-FE-db	1.17	0.202	L	L	L	L	0	L	L
LRapw-FE-db	1.16	0.247	0	L	L	L	L	0	L
LRapw-FE-db	1.16	0.142	0	0	0	L	S	0	L
MLM-FE	1.34	0.0638	L	L	0	S	0	L	L
MLM-FE	1.32	0.0996	L	L	0	L	L	L	0
MLM-FE	1.31	0.0888	L	L	0	L	S	L	L
MLM-FE	1.31	0.0941	L	L	0	L	0	L	L
MLM-FE	1.31	0.1	L	L	0	L	S	L	0
MLM-FIcw	1.48	0.1	L	L	0	L	S	L	L
MLM-FIcw	1.39	0.0999	L	L	0	L	0	L	L
MLM-FIcw	1.33	0.101	L	L	0	L	L	L	0
MLM-FIcw	1.32	0.101	L	L	0	L	S	L	0
MLM-FIcw	1.32	0.101	L	L	0	L	0	L	0
MLM-RE	1.35	0.0646	L	L	0	S	0	L	L
MLM-RE	1.34	0.0963	L	L	0	L	0	L	L
MLM-RE	1.32	0.0996	L	L	0	L	L	L	0
MLM-RE	1.32	0.0891	L	L	0	L	S	L	L
MLM-RE	1.31	0.1	L	L	0	L	S	L	0
LRa-FIcw-db	1.41	0.0635	L	0	0	L	S	0	0

method	RMSE.s.estimand.norm	RMSE.s estimand	J.bar	sd school	ICC school	cv school	num.cov school	size.impact cor.school	block type
LRa-FIcw-db	1.4	0.0632	L	0	0	L	0	L	0
LRa-FIcw-db	1.4	0.0635	L	0	0	L	S	L	0
LRa-FIcw-db	1.4	0.0633	L	0	0	L	L	L	0
LRa-FIcw-db	1.4	0.0811	S	0	0	L	0	L	0
LRapw-FIpw-db	1.13	0.112	L	0	L	L	L	0	L
LRapw-FIpw-db	1.12	0.177	L	0	L	L	S	0	L
LRapw-FIpw-db	1.11	0.238	S	L	L	L	L	L	L

The Effect of Different Data Generating Parameters on the Estimate Instability

Term	LRa-FE-db	LRa-FIcw-db	LRapw-FE-db	LRapw-FIpw-db	MLM-FE	MLM-FIcw	MLM-RE
(Intercept)	.067***	.067***	.069***	.070***	.068***	.069***	.068***
block_typeS	.002	.002	.002	.002	.002	.002	.002
block_typeL	<.001	<.001	-.001	<.001	.001	<.001	.001
J.barS	-.011***	-.011***	-.010**	-.010**	-.011***	-.011***	-.011***
J.barL	-.024***	-.024***	-.022***	-.022***	-.024***	-.024***	-.023***
size.impact.cor.schoolL	-.003	-.003	-.002	-.003	-.002	-.003	-.002
sd.schoolL	.026***	.026***	.027***	.026***	.026***	.026***	.026***
ICC.schoolL	.094***	.094***	.091***	.090***	.093***	.092***	.093***
variable.nTRUE	.005.	.005.	-.003	-.003	.001	.001	.001
cv.schoolS	.001	.001	-.001	-.001	<.001	<.001	<.001
cv.schoolL	.004	.004	-.006.	-.005	-.001	-.001	-.001
num.cov.schoolS	-.001	-.001	-.001	-.001	-.001	-.001	-.001
num.cov.schoolL	<.001	<.001	<.001	<.001	<.001	<.001	<.001
block_typeS:J.barS	-.002	-.002	-.002	-.002	-.002	-.002	-.002
block_typeL:J.barS	.004	.004.	.003	.003	.003	.004	.003
block_typeS:J.barL	-.001	-.001	-.001	-.001	-.001	-.001	-.001
block_typeL:J.barL	.003	.003	.001	.002	.002	.002	.002
block_typeS: size.impact.cor.schoolL	<.001	<.001	<.001	<.001	<.001	<.001	<.001
block_typeL: size.impact.cor.schoolL	.002	.002	.002	.003	.002	.002	.002
block_typeS:sd.schoolL	.001	.001	.001	.001	.001	.001	.001
block_typeL:sd.schoolL	.003	.002	.001	.001	.002	.002	.002
block_typeS:ICC.schoolL	<.001	<.001	<.001	<.001	<.001	<.001	<.001
block_typeL:ICC.schoolL	.008***	.009***	.006***	.008***	.007***	.009***	.006***
block_typeS: variable.nTRUE	.001	.001	.001	.001	.001	.001	.001
block_typeL: variable.nTRUE	.002	.002	.008***	.005**	.003.	.002	.003.
block_typeS:cv.schoolS	-.001	-.001	<.001	<.001	-.001	-.001	<.001
block_typeL:cv.schoolS	.002	.002	.004.	.004.	.002	.002	.002
block_typeS:cv.schoolL	-.001	-.001	-.001	-.001	-.001	-.001	-.001
block_typeL:cv.schoolL	.004.	.004.	.011***	.008***	.006**	.004.	.006**
block_typeS: num.cov.schoolS	.001	.001	.001	.001	.001	.001	.001

Term	LRa-FE-db	LRa-FIcw-db	LRapw-FE-db	LRapw-FIpw-db	MLM-FE	MLM-FIcw	MLM-RE
block_typeL: num.cov.schoolS	-.001	-.001	-.001	-.001	-.001	-.001	-.001
block_typeS: num.cov.schoolL	.001	.001	.001	.001	.001	.001	.001
block_typeL: num.cov.schoolL	-.002	-.002	-.002	-.002	-.002	-.002	-.002
J.barS: size.impact.cor.schoolL	<.001	<.001	<.001	<.001	<.001	<.001	<.001
J.barL: size.impact.cor.schoolL	-.003	-.003	-.003	-.003	-.003	-.003	-.003
J.barS:sd.schoolL	-.002	-.002	-.002	-.002	-.002	-.002	-.002
J.barL:sd.schoolL	-.003.	-.003.	-.003.	-.003.	-.003.	-.004.	-.003.
J.barS:ICC.schoolL	-.015***	-.015***	-.016***	-.016***	-.015***	-.015***	-.015***
J.barL:ICC.schoolL	-.029***	-.028***	-.030***	-.030***	-.029***	-.029***	-.029***
J.barS:variable.nTRUE	<.001	<.001	<.001	<.001	<.001	<.001	<.001
J.barL:variable.nTRUE	-.004*	-.004.	-.005*	-.004*	-.003	-.003	-.003
J.barS:cv.schoolS	<.001	<.001	<.001	<.001	<.001	<.001	<.001
J.barL:cv.schoolS	<.001	<.001	<.001	<.001	<.001	<.001	<.001
J.barS:cv.schoolL	-.003	-.003	-.004	-.003	-.003	-.003	-.003
J.barL:cv.schoolL	-.003	-.003	-.004.	-.003	-.002	-.002	-.002
J.barS:num.cov.schoolS	<.001	<.001	<.001	<.001	<.001	<.001	<.001
J.barL:num.cov.schoolS	.002	.002	.002	.003	.002	.002	.002
J.barS:num.cov.schoolL	-.001	-.001	-.001	-.001	-.001	-.001	-.001
J.barL:num.cov.schoolL	-.001	-.001	-.002	-.001	-.001	-.001	-.001
size.impact.cor.schoolL: sd.schoolL	-.001	-.001	-.002	-.001	-.001	-.001	-.001
size.impact.cor.schoolL: ICC.schoolL	.002	.002	.003.	.003.	.002	.002	.002
size.impact.cor.schoolL: variable.nTRUE	.003.	.003.	.002	.003.	.003.	.003.	.003.
size.impact.cor.schoolL: cv.schoolS	<.001	<.001	<.001	<.001	<.001	<.001	<.001
size.impact.cor.schoolL: cv.schoolL	.001	.001	.001	.001	.001	.001	.001
size.impact.cor.schoolL: num.cov.schoolS	.002	.002	.002	.002	.002	.002	.002
size.impact.cor.schoolL: num.cov.schoolL	.002	.002	.002	.002	.002	.002	.002
sd.schoolL:ICC.schoolL	-.010***	-.010***	-.011***	-.011***	-.011***	-.011***	-.011***
sd.schoolL:variable.nTRUE	-.003.	-.003.	-.002	-.001	-.002	-.001	-.002
sd.schoolL:cv.schoolS	-.004*	-.004*	-.004*	-.004.	-.004*	-.004*	-.004*
sd.schoolL:cv.schoolL	-.003.	-.003	-.001	-.001	-.002	-.001	-.002
sd.schoolL: num.cov.schoolS	<.001	<.001	-.001	-.001	-.001	-.001	-.001
sd.schoolL: num.cov.schoolL	-.001	-.001	-.001	-.001	-.001	-.001	-.001
ICC.schoolL: variable.nTRUE	-.003.	-.003.	.008***	.008***	.002	.003.	.002
ICC.schoolL:cv.schoolS	-.001	-.001	.002	.002	<.001	<.001	<.001
ICC.schoolL:cv.schoolL	-.006**	-.006**	.007***	.007***	<.001	.001	<.001
ICC.schoolL: num.cov.schoolS	-.002	-.002	-.002	-.002	-.002	-.002	-.002
ICC.schoolL: num.cov.schoolL	-.001	-.001	-.002	-.002	-.001	-.001	-.001
variable.nTRUE:cv.schoolS	.001	.001	.004*	.003.	<.001	<.001	<.001
variable.nTRUE:cv.schoolL	.015***	.015***	.016***	.014***	.005**	.004*	.005**
variable.nTRUE: num.cov.schoolS	-.004*	-.004*	-.004*	-.004*	-.004*	-.004*	-.004*
variable.nTRUE: num.cov.schoolL	-.001	-.001	-.001	-.001	-.001	-.001	-.001
cv.schoolS: num.cov.schoolS	.004.	.005*	.004.	.005*	.004.	.005*	.004*
cv.schoolL: num.cov.schoolS	.005*	.005*	.005*	.005*	.005*	.005*	.005*
cv.schoolS: num.cov.schoolL	.001	.001	.001	.001	.001	.001	.001

Term	LRa-FE-db	LRa-FIcw-db	LRapw-FE-db	LRapw-FIpw-db	MLM-FE	MLM-FIcw	MLM-RE
cv.schoolL: num.cov.schoolL	.002	.002	.002	.002	.002	.002	.002

*Significance: '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1*

Top Five Scenarios with Largest Normalized Estimate Standard Error by Method

method	SE s.norm	SE.s	J.bar	sd school	ICC school	cv school	num.cov school	size.impact cor.school	block type
LRa-FE-db	1.41	0.0635	L	0	0	L	S	0	0
LRa-FE-db	1.4	0.0632	L	0	0	L	0	L	0
LRa-FE-db	1.4	0.0635	L	0	0	L	S	L	0
LRa-FE-db	1.4	0.0633	L	0	0	L	L	L	0
LRa-FE-db	1.4	0.0811	S	0	0	L	0	L	0
LRapw-FE-db	1.21	0.0918	L	0	0	L	S	0	L
LRapw-FE-db	1.19	0.139	0	0	0	L	L	0	L
LRapw-FE-db	1.19	0.0731	S	0	0	S	L	0	L
LRapw-FE-db	1.17	0.184	L	L	L	L	0	L	L
LRapw-FE-db	1.15	0.222	0	0	L	L	L	0	L
LRa-FIcw-db	1.41	0.0635	L	0	0	L	S	0	0
LRa-FIcw-db	1.4	0.0632	L	0	0	L	0	L	0
LRa-FIcw-db	1.4	0.0635	L	0	0	L	S	L	0
LRa-FIcw-db	1.4	0.0633	L	0	0	L	L	L	0
LRa-FIcw-db	1.4	0.0811	S	0	0	L	0	L	0
LRapw-FIpw-db	1.13	0.112	L	0	L	L	L	0	L
LRapw-FIpw-db	1.12	0.134	L	0	L	L	0	L	L
LRapw-FIpw-db	1.12	0.177	L	0	L	L	S	0	L
LRapw-FIpw-db	1.11	0.213	0	0	L	L	L	0	L
LRapw-FIpw-db	1.11	0.123	L	0	L	L	L	L	L

What do the estimated standard errors look like for each estimator

Main Bullet Points:

*This section answers the rest of the questions from pg.39 (i.e. are certain SE estimators more accurate than others)

*For the cluster targeted estimates the unbiased estimators produce more accurate standard errors than the biased estimators (graph one). The unbiased estimators produce more unstable estimates than the biased estimators, but in 90% of the DGP we looked at they aren't a lot more unstable (graph four).

*For the person targeted estimates the unbiased estimators produce more accurate standard errors than the biased estimators (graph one), except when the school ICC is zero and the blocks are all very dissimilar from each other and their is high variance in the cluster sizes. if all three of these risk factors are not present than the unbiased estimator standard estimates are generally fine (graph two). The unbiased estimators produce more unstable estimates than the biased estimators unless all three of these factors are present (graphs four and five). However they aren't a lot more unstable, 90% of the time they're within about 10% of the median instability.

*This section also answers the question from pg. 38 about the instability of the estimated standard erros for the fully interacted models, what I've been calling hte unbiased estimators. The simulations do not really show that the standard errors from the fully interacted models are lot more unstable than the other standard errors. We should talk about this more, and what would be helpful for me to dig into more.

Description of Analysis:

*The first graph is a boxplot of the estimated standard error accuracy across the different scenarios by method. Standard error accuracy is quantified as the estimated standard error over the true super population standard error. Estimated standard errors are most accurate if the number is close to one. Numbers below one represent under estimated standard errors and numbers above one represent over estimated standard errors.

*The second graph is the same as the first but for just the person weighted fully interacted estimators broken out by identified "risk factors" for poorly estimated standard errors. In the graph on the left the school level ICC is zero, the blocks are all messy, and there is high variance in the cluster size and the graph on the right does not have all three of these things at the same time.

*The third graph is a boxplot of the standard deviation of the estimated standard error across the different replications by method.

*The forth graph is a boxplot of the normalized standard deviation of the estimated standard error across the different replications by method.

*The fifth graph is the same as the fourth graph but broken by the same risk factors as before.

*The second graph is a boxplot of the normalized estimate standard deviation across the different scenarios by method. This is also normalized by dividing by the median estimate standard deviation within a given simulation scenario.

*The first table contains the regression coefficients from a regression that predicts the RMSE using different data generating parameters and the second table is from regressions predicting the estimate standard deviation. Each regression is run separately by estimator and the RMSE is against the targeted estimator.

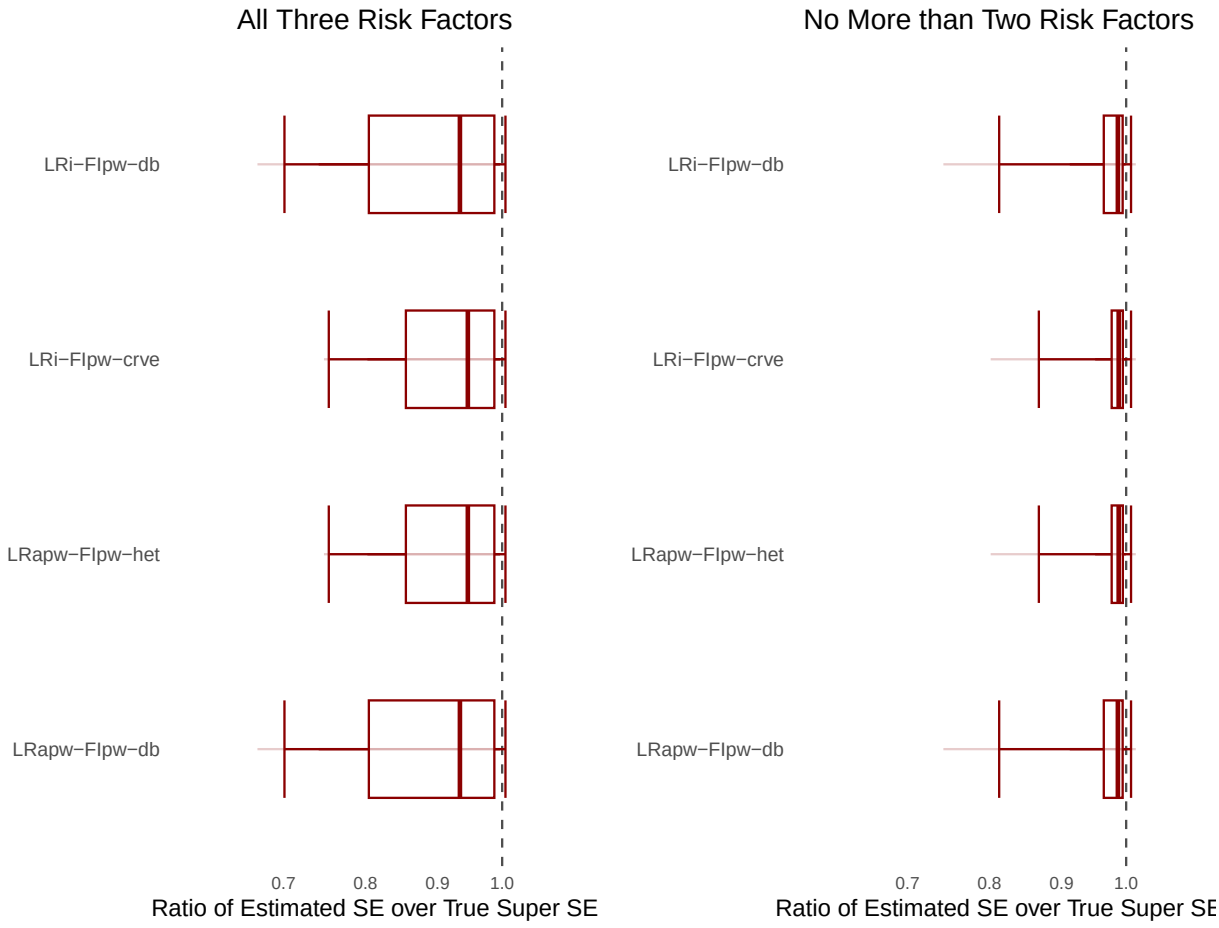
*The second table contains the DGPs that produce the normalized RMSE (against the targeted estimator) for each method, conditional on the magnitude of the being at least 1.10.

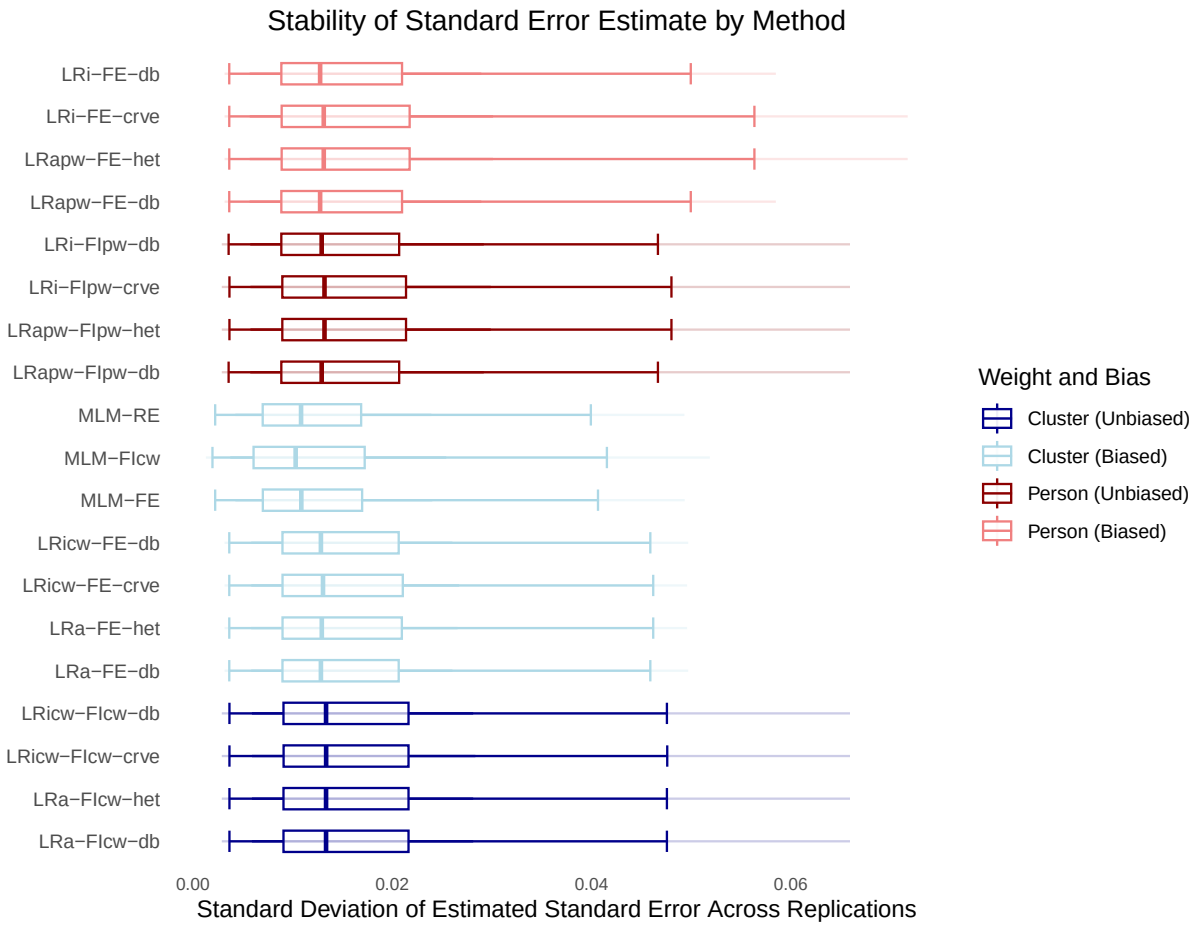
*The third table contains the regression coefficients from a regression that predicts the true superpopulation estimate standard error using different data generating parameters.

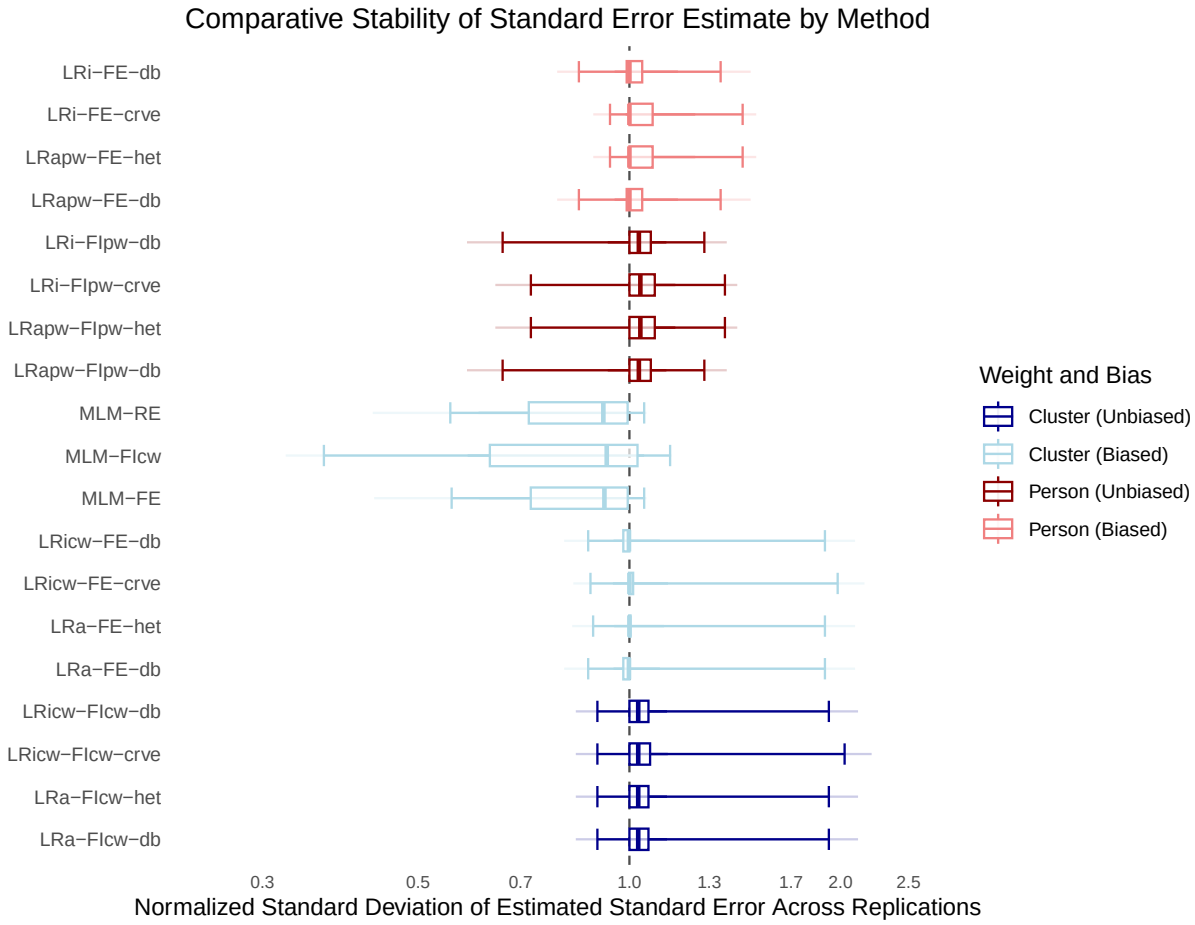
*The fourth table contains the DGPs that produce the normalized super population estimator standard error for each method, conditional on the magnitude of the being at least 1.10.


Super Population Estimated Standard Error Accuracy by Method
LRi−FE−db
LRi−FE−crve
LRapw−FE−het
LRapw−FE−db
LRi−FIpw−db
LRi−FIpw−crve
LRapw−FIpw−het
LRapw−FIpw−db
MLM−RE
MLM−FIcw
MLM−FE
LRicw−FE−db
LRicw−FE−crve
LRa−FE−het
LRa−FE−db
LRicw−FIcw−db
LRicw−FIcw−crve
LRa−FIcw−het
LRa−FIcw−db
0.7
1.0
1.3
1.7
2.0
2.5
Ratio of Estimated SE over True Super SE
Weight and Bias
Cluster (Unbiased)
Cluster (Biased)
Person (Unbiased)
Person (Biased)

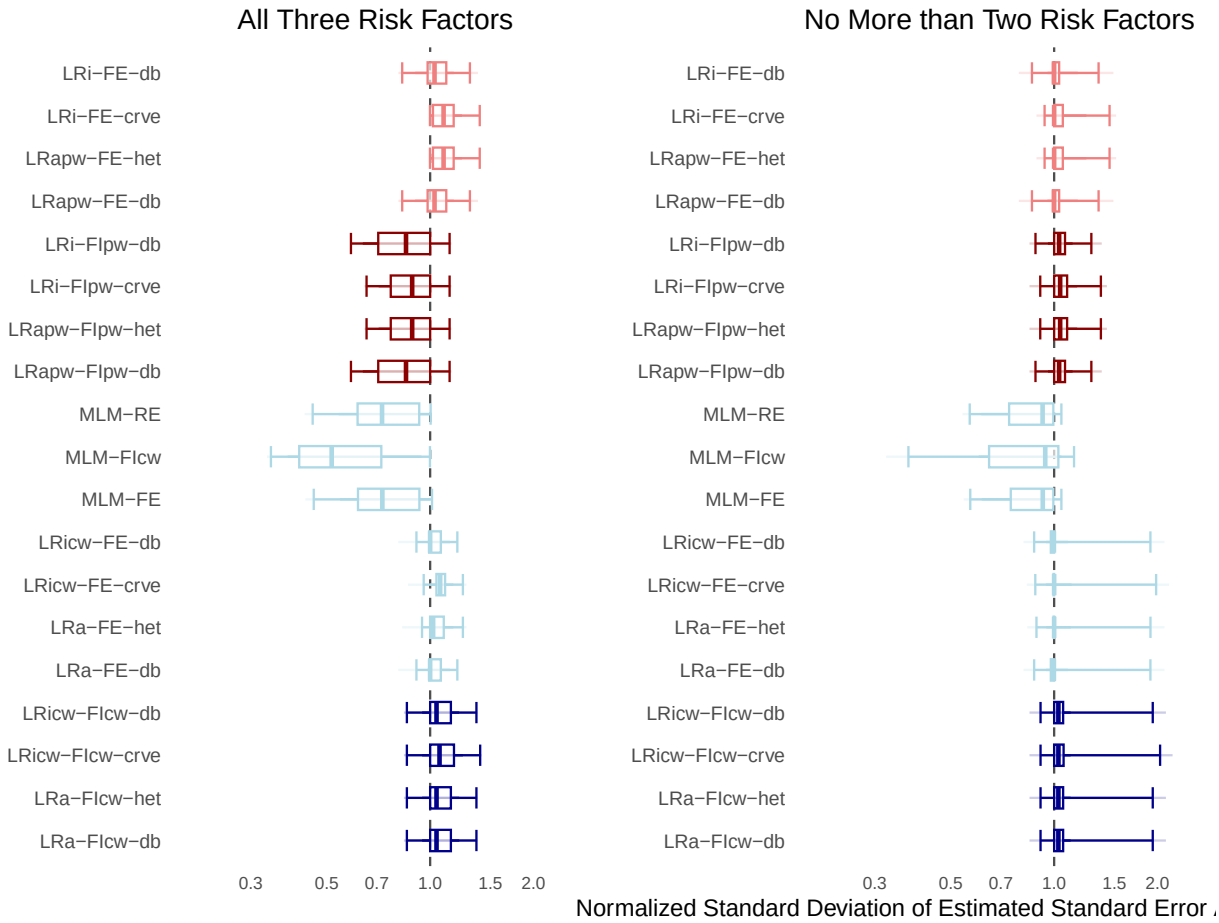
The Risk Factors are School Level ICC = 0 & Blocks All Different & High Variance in Cluster Size




Stability of Standard Error Estimate by Method
LRi−FE−db
LRi−FE−crve
LRapw−FE−het
LRapw−FE−db
LRi−FIpw−db
LRi−FIpw−crve
LRapw−FIpw−het
LRapw−FIpw−db
MLM−RE
MLM−FIcw
MLM−FE
LRicw−FE−db
LRicw−FE−crve
LRa−FE−het
LRa−FE−db
LRicw−FIcw−db
LRicw−FIcw−crve
LRa−FIcw−het
LRa−FIcw−db
0.00
0.02
0.04
0.06
Standard Deviation of Estimated Standard Error Across Replications
Weight and Bias
Cluster (Unbiased)
Cluster (Biased)
Person (Unbiased)
Person (Biased)


Comparative Stability of Standard Error Estimate by Method
LRi−FE−db
LRi−FE−crve
LRapw−FE−het
LRapw−FE−db
LRi−FIpw−db
LRi−FIpw−crve
LRapw−FIpw−het
LRapw−FIpw−db
MLM−RE
MLM−FIcw
MLM−FE
LRicw−FE−db
LRicw−FE−crve
LRa−FE−het
LRa−FE−db
LRicw−FIcw−db
LRicw−FIcw−crve
LRa−FIcw−het
LRa−FIcw−db
0.3
0.5
0.7
1.0
1.3
1.7
2.0
2.5
Normalized Standard Deviation of Estimated Standard Error Across Replications
Weight and Bias
Cluster (Unbiased)
Cluster (Biased)
Person (Unbiased)
Person (Biased)

The Risk Factors are School Level ICC = 0 & Blocks All Different & High Variance in Cluster Size



The Effect of Different Data Generating Parameters on % Points Estimated SE Off from True Super SE

Term	LRa-FE-db	LRa-FE-het	LRa-FIcw-db	LRapw-FE-db	LRapw-FE-het	LRapw-FIpw-db	LRapw-FIpw-het	LRicw-FE-crve	LRicw-FIcw-crve	MLM-FE	MLM-FIcw	MLM-RE
(Intercept)	5.842**	6.070**	1.069***	5.501**	5.975**	.990*	1.092**	5.679**	1.217***	8.017***	5.468***	8.038***
block_typeS	.785	.885	.288.	.794	1.200	.146	.145	.947	.313.	.157	-.929.	.170
block_typeL	79.662***	82.928***	.755***	81.702***	84.100***	-.307	-.166	84.159***	.977***	57.767***	1.269*	57.528***
J.barS	-.501	-.572	-.410*	-.366	-.595	.243	.145	-.525	-.495**	-1.450	-1.872***	-1.429
J.barL	-.546	-.854	-.676***	-.829	-.887	.908*	.536	-.905	-.780***	-3.217*	-3.597***	-3.246*
size.impact.cor.schoolL	-2.549	-2.439	-.060	-3.061.	-2.992*	-.578	-.517	-2.456	.009	-1.619	-.292	-1.614
sd.schoolL	-5.786***	-5.976***	-.041	-6.463***	-6.511***	.074	.182	-5.977***	-.071	-4.594***	.238	-4.654***
ICC.schoolL	-9.989***	-10.379***	-.105	-9.942***	-9.782***	.849*	.908**	-10.238***	-.391**	-10.638***	4.002***	-10.585***
variable.nTRU E	-.076	-.245	.109	1.458	.164	-.769*	-1.147***	.289	-.309*	1.030	.138	1.063
cv.schoolS	.132	-.093	.211	.784	.259	-.079	-.144	.031	.144	.733	.020	.639
cv.schoolL	-.230	-.296	.599***	1.418	-.034	-.427	-.929*	.381	.040	-.510	-.315	-.555
num.cov.schoo lS	-1.140	-1.279	.017	-1.376	-1.657	-.011	-.005	-1.286	-.001	-.418	.024	-.432
num.cov.schoo lL	.135	.174	.013	-.072	-.183	-.027	-.085	.250	.082	-.201	-.084	-.234
block_typeS: J.barS	.250	.183	-.103	.153	.079	-.169	-.196	.138	-.074	.985	.919*	.978
block_typeL: J.barS	-1.105	-1.001	-.107	-.396	-.998	.089	-.178	-.756	-.059	-1.544	1.624***	-1.557
block_typeS: J.barL	.422	.361	-.096	.338	.271	-.102	-.199	.383	.039	.967	.993*	.959
block_typeL: J.barL	-2.036	-2.360	-.462***	.170	-1.841	-.569.	-.907**	-1.648	-.217.	-3.667***	1.827***	-3.489**
block_typeS: size.impact.cor.schoolL	.021	.025	-.022	.021	-.020	.023	.028	.035	-.044	.069	.113	.076
block_typeL: size.impact.cor.schoolL	-2.313.	-2.316*	.126	-2.014.	-1.956.	.408	.378	-2.270.	.077	-.906	-.793*	-.822
block_typeS: sd.schoolL	-.281	-.337	.016	-.090	-.614	.062	.052	-.309	.016	-1.299	-.543.	-1.294
block_typeL: sd.schoolL	-17.243***	-17.420***	-.024	-16.305***	-16.905***	-.988***	-.910***	-17.414***	-.059	-10.763***	3.666***	-10.700***

Term	LRa-FE-db	LRa-FE-het	LRa-FIcw-db	LRapw-FE-db	LRapw-FE-het	LRapw-FIpw-db	LRapw-FIpw-het	LRicw-FE-crve	LRicw-FIcw-crve	MLM-FE	MLM-FIcw	MLM-RE
block_typeS: ICC.schoolL	-.614	-.633	.002	-.344	-.851	-.200	-.135	-.645	-.028	-.271	.646*	-.266
block_typeL: ICC.schoolL	-47.760***	-49.338***	-.140	-47.155***	-49.943***	-1.918***	-1.798***	-49.892***	-.370***	-34.502***	-2.556***	-34.672***
block_typeS: variable.nTRUE	-.383	-.383	-.036	-.701	-.299	.357	.350	-.471	-.110	-.143	.016	-.148
block_typeL: variable.nTRUE	-5.922***	-6.151***	.138	-13.397***	-8.641***	4.563***	4.528***	-8.710***	-.217*	-3.898***	-.235	-3.815***
block_typeS: cv.schoolS	-.171	-.165	-.191	-.501	-.250	-.010	-.033	-.224	-.217.	.018	-.156	-.003
block_typeL: cv.schoolS	-.005	-.016	.043	-4.063**	-2.210.	2.014***	1.957***	-.585	-.024	.123	-.448	.114
block_typeS: cv.schoolL	-.449	-.458	-.217.	-.596	-.248	.192	.205	-.532	-.302*	.185	.238	.195
block_typeL: cv.schoolL	-6.736***	-6.960***	.067	-13.890***	-8.502***	4.734***	4.738***	-10.231***	-.398**	-3.676***	-.357	-3.635**
block_typeS: num.cov.schoolS	-.149	-.146	-.047	-.102	-.139	.109	.101	-.154	-.062	-.146	-.173	-.153
block_typeL: num.cov.schoolS	-1.002	-1.007	-.025	-1.112	-1.155	-.075	-.051	-.997	.028	-.187	.039	-.127
block_typeS: num.cov.schoolL	.033	.032	.071	-.074	-.044	-.008	.046	-.029	.037	.020	-.003	.017
block_typeL: num.cov.schoolL	.687	.563	.141	.211	.003	.339	.298	.495	.132	-1.117	-.615	-1.080
J.barS: size.impact.cor.schoolL	.592	.661	.078	.571	.656	.152	.113	.682	.095	.243	.324	.212
J.barL: size.impact.cor.schoolL	<.001	.117	.011	.208	.217	-.110	-.094	.177	.052	-.364	.404	-.369
J.barS: sd.schoolL	.224	.273	.131	.024	.294	.157	.116	.212	.150	-1.172	-.547.	-1.243
J.barL: sd.schoolL	.921	1.003	.100	.563	.915	.221	.130	.899	.129	.266	.031	.225
J.barS: ICC.schoolL	.264	.349	.126	-.003	.202	-.180	-.263	.213	.186.	1.470	.992**	1.464
J.barL: ICC.schoolL	1.069	1.189	.282**	.304	.638	-.311	-.380.	.932	.258*	2.851**	1.347***	2.827**
J.barS: variable.nTRUE	1.003	1.049	.057	.728	.945	-1.187***	-.666**	.982	.132	-.344	-.035	-.342
J.barL: variable.nTRUE	-.825	-.623	-.046	-.844	-.530	-2.689***	-1.491***	-.728	-.139	-.577	.729*	-.588
J.barS: cv.schoolS	-.735	-.514	-.003	-1.145	-.949	-.490	-.289	-.542	-.018	-.905	-.113	-.883
J.barL: cv.schoolS	.316	.432	.003	.199	.280	-1.051***	-.578*	.394	-.112	.937	.413	1.056
J.barS: cv.schoolL	-1.044	-1.113	-.328**	-1.047	-.834	-1.707***	-1.126***	-1.186	-.200	-.012	.246	-.008
J.barL: cv.schoolL	-.871	-.903	-.366**	-.782	-.613	-3.277***	-1.953***	-1.023	-.390**	.435	.183	.508
J.barS: num.cov.schoolS	-.088	-.044	-.022	.118	.303	-.033	-.006	-.020	-.057	.245	-.264	.257
J.barL: num.cov.schoolS	.702	.866	.052	.515	.826	.310	.254	.840	.052	.623	.345	.611
J.barS: num.cov.schoolL	-.500	-.606	-.219.	-.579	-.430	-.122	-.100	-.605	-.259*	.635	-.290	.711
J.barL: num.cov.schoolL	-1.613	-1.466	-.122	-1.298	-1.085	-.166	-.139	-1.412	-.121	-.710	-.048	-.663
size.impact.cor.schoolL: sd.schoolL	1.442	1.268	.029	1.808.	1.551.	.336	.281	1.306	.005	.271	-.332	.303
size.impact.cor.schoolL: ICC.schoolL	1.866.	1.821.	.059	1.332	1.446.	.197	.167	1.727.	.024	.687	.537*	.641
size.impact.cor.schoolL: variable.nTRUE	.302	.238	.196*	.750	.658	.658**	.588**	.269	.136	.295	-.185	.287
size.impact.cor.schoolL: cv.schoolS	2.144.	2.058.	-.160.	2.125.	2.088.	.035	-.008	2.041.	-.177.	1.714.	.243	1.707.
size.impact.cor.schoolL: cv.schoolL	1.479	1.430	-.066	2.170.	2.032.	.430.	.369	1.495	-.140	1.760.	.505	1.766.
size.impact.cor.schoolL: num.cov.schoolS	-1.287	-1.223	.005	-1.046	-.909	-.058	-.014	-1.233	.007	-.836	-.330	-.805
size.impact.cor.schoolL: num.cov.schoolL	-.632	-.660	-.091	-.243	-.218	-.113	-.059	-.699	-.110	.551	-.221	.571
sd.schoolL: ICC.schoolL	9.881***	10.098***	.112	9.106***	9.813***	-.117	-.063	9.915***	-.059	8.586***	-.285	8.563***
sd.schoolL: variable.nTRUE	.529	.677	-.171*	1.716.	1.278	.093	-.046	.910	.091	.513	-.037	.551
sd.schoolL: cv.schoolS	-2.729*	-2.534*	-.085	-1.845	-1.948.	-.094	-.167	-2.455*	.008	-1.367	-.384	-1.271
sd.schoolL: cv.schoolL	-.079	.002	-.105	.818	.318	.301	.164	.274	.196.	.439	-.139	.497
sd.schoolL: num.cov.schoolS	1.332	1.318	.017	1.329	1.496	.007	-.047	1.298	<.001	.506	.175	.551
sd.schoolL: num.cov.schoolL	-.515	-.535	.017	-.220	-.177	-.064	-.058	-.514	.008	-.597	-.227	-.564
ICC.schoolL: variable.nTRUE	4.219***	4.340***	-.196*	5.149***	4.426***	.073	-.092	4.945***	.736***	2.511***	.120	2.483***
ICC.schoolL: cv.schoolS	.260	.423	-.126	1.464	1.186	-.133	-.289	.582	.085	-.177	.366	-.175
ICC.schoolL: cv.schoolL	3.902**	4.208***	-.396***	4.093***	3.573***	.014	-.076	4.956***	.791***	2.307*	.248	2.294*
ICC.schoolL: num.cov.schoolS	.304	.340	-.028	.614	.599	-.196	-.201	.380	-.008	-.124	-.072	-.111
ICC.schoolL: num.cov.schoolL	-.217	-.213	-.015	.056	.119	-.095	-.089	-.207	-.041	.447	.481	.477
variable.nTRUE: cv.schoolS	-1.044	-1.117	.025	-2.644*	-2.246*	3.750***	2.253***	-1.277	.360***	-1.640.	-.550.	-1.635.
variable.nTRUE: cv.schoolL	-5.616***	-5.661***	.568***	-6.344***	-5.707***	10.657***	6.163***	-5.877***	2.479***	-4.975***	-.499	-4.965***
variable.nTRUE: num.cov.schoolS	.188	.204	-.098	.141	.099	-.036	-.027	.209	-.070	.551	.231	.521
variable.nTRUE: num.cov.schoolL	-.281	-.241	.109	-.594	-.606	.256	.266	-.331	.079	-.661	.268	-.749
cv.schoolS: num.cov.schoolS	.342	.437	.155	.440	.470	.135	.145	.442	.158	-.602	.024	-.606
cv.schoolL: num.cov.schoolS	1.890	1.844	.041	1.722	1.653	.154	.158	1.846	.081	1.122	.073	1.086
cv.schoolS: num.cov.schoolL	1.944	1.886	.112	1.703	1.594	.122	.123	1.872	.105	.315	-.158	.311
cv.schoolL: num.cov.schoolL	2.079	1.980	-.048	1.849	1.724	.162	.176	1.858	-.086	.719	.262	.709

*Significance: '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1*

Top Five Scenarios with Most SE Estimate Inaccuracy by Method

method	infla te.s	J.bar	sd school	ICC school	cv school	num.cov school	size.impact cor.school	block type
LRa-FE-db	2.53	S	0	0	S	L	0	L
LRa-FE-db	2.39	0	0	0	0	0	0	L
LRa-FE-db	2.28	0	0	0	L	L	0	L
LRa-FE-db	2.19	S	0	0	L	L	0	L
LRa-FE-db	2.13	0	0	0	S	L	L	L
LRa-FE-het	2.55	S	0	0	S	L	0	L
LRa-FE-het	2.39	0	0	0	0	0	0	L
LRa-FE-het	2.32	0	0	0	L	L	0	L
LRa-FE-het	2.21	S	0	0	L	L	0	L
LRa-FE-het	2.17	0	0	0	S	L	0	L
LRapw-FE-db	2.39	0	0	0	0	0	0	L
LRapw-FE-db	2.28	0	0	0	L	L	0	L
LRapw-FE-db	2.19	S	0	0	L	L	0	L
LRapw-FE-db	2.14	S	0	0	S	L	0	L
LRapw-FE-db	2.13	0	0	0	S	L	0	L
LRapw-FE-het	2.39	0	0	0	0	0	0	L
LRapw-FE-het	2.32	0	0	0	L	L	0	L
LRapw-FE-het	2.21	S	0	0	L	L	0	L
LRapw-FE-het	2.19	S	0	0	S	L	0	L
LRapw-FE-het	2.17	0	0	0	S	L	0	L
LRicw-FE-crve	2.54	S	0	0	S	L	0	L
LRicw-FE-crve	2.39	0	0	0	0	0	0	L
LRicw-FE-crve	2.32	0	0	0	L	L	0	L
LRicw-FE-crve	2.21	S	0	0	L	L	0	L

method	infla te.s	J.bar	sd school	ICC school	cv school	num.cov school	size.impact cor.school	block type
LRicw-FE-crve	2.17	0	0	0	S	L	0	L
MLM-FE	2.02	L	0	0	S	0	0	L
MLM-FE	1.97	S	0	0	L	S	L	L
MLM-FE	1.9	S	0	0	0	L	0	L
MLM-FE	1.89	L	0	0	0	S	0	L
MLM-FE	1.89	L	0	0	S	L	L	L
MLM-FIcw	1.27	L	L	0	0	S	0	L
MLM-FIcw	1.25	0	L	0	0	0	0	L
MLM-FIcw	1.25	L	L	0	0	L	0	L
MLM-FIcw	1.25	L	L	0	S	S	L	L
MLM-FIcw	1.25	L	L	0	S	0	L	L
MLM-RE	2.03	L	0	0	S	0	0	L
MLM-RE	1.97	S	0	0	L	S	L	L
MLM-RE	1.9	S	0	0	0	L	0	L
MLM-RE	1.89	L	0	0	0	S	0	L
MLM-RE	1.89	L	0	0	S	L	L	L
LRapw-FIpw-db	0.671	0	0	0	L	L	0	L
LRapw-FIpw-db	0.714	0	0	0	L	0	L	L
LRapw-FIpw-db	0.714	0	0	0	L	S	0	L
LRapw-FIpw-db	0.716	S	0	0	L	L	L	L
LRapw-FIpw-db	0.723	L	0	0	L	S	0	L
LRapw-FIpw-het	0.748	0	0	0	L	L	0	L
LRapw-FIpw-het	0.756	S	0	0	L	L	L	L
LRapw-FIpw-het	0.761	L	0	0	L	S	0	L
LRapw-FIpw-het	0.767	0	0	0	L	S	0	L
LRapw-FIpw-het	0.781	0	0	0	L	S	L	L

The Effect of Different Data Generating Parameters on the Estimate Standard Error Stability

Term	LRa-FE-db	LRa-FE-het	LRa-FIcw-db	LRapw-FE-db	LRapw-FE-het	LRapw-FIpw-db	LRapw-FIpw-het	LRicw-FE-crve	LRicw-FIcw-crve	MLM-FE	MLM-FIcw	MLM-RE
(Intercept)	.009***	.009***	.010***	.010***	.010***	.010***	.010***	.009***	.010***	.006***	.006***	.006***
block_typeS	.001	.001	.001	.001	.001	.001	.001	.001	.001	.001	<.001	.001
block_typeL	.006***	.006***	.003**	.005***	.005***	.004***	.004**	.006***	.003**	.006***	.001	.006***
J.barS	-.003**	-.003**	-.003**	-.003*	-.002.	-.003**	-.003**	-.003**	-.003**	-.002*	-.002*	-.002*
J.barL	-.005***	-.005***	-.006***	-.004***	-.003**	-.006***	-.006***	-.005***	-.006***	-.004***	-.004***	-.004***

Term	LRa-FE-db	LRa-FE-het	LRa-FIcw-db	LRapw-FE-db	LRapw-FE-het	LRapw-FIpw-db	LRapw-FIpw-het	LRicw-FE-crve	LRicw-FIcw-crve	MLM-FE	MLM-FIcw	MLM-RE
size.impact.cor.schoolL	-.001	-.001	-.001	-.001	-.001	-.001	-.001	-.001	-.001	-.001	-.001	-.001
sd.schoolL	.004***	.004***	.005***	.005***	.004***	.005***	.005***	.004***	.005***	.007***	.009***	.007***
ICC.schoolL	.012***	.012***	.014***	.011***	.011***	.013***	.013***	.012***	.014***	.015***	.017***	.015***
variable.nTRU E	.002*	.002*	.002.	-.001	-.001	<.001	<.001	.002*	.002*	.001	.001	.001
cv.schoolS	.001	<.001	<.001	<.001	<.001	<.001	<.001	.001	<.001	<.001	<.001	<.001
cv.schoolL	.002*	.002.	.002.	-.002.	-.002.	-.001	-.001	.002*	.002*	<.001	<.001	<.001
num.cov.schoo lS	-.001	-.001	-.001	<.001	<.001	-.001	-.001	-.001	-.001	<.001	-.001	<.001
num.cov.schoo lL	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001
block_typeS: J.barS	-.001	-.001	-.001	-.001	-.001	-.001	-.001	-.001	-.001	<.001	<.001	<.001
block_typeL: J.barS	<.001	<.001	.001	<.001	-.001	<.001	.001	<.001	.001	-.001	<.001	-.001
block_typeS: J.barL	-.001	-.001	-.001	-.001	-.001	-.001	-.001	-.001	-.001	<.001	<.001	<.001
block_typeL: J.barL	-.003***	-.003***	-.002*	-.004***	-.004***	-.002*	-.001.	-.003***	-.002.	-.003***	-.001	-.003***
block_typeS: size.impact.cor.schoolL	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001
block_typeL: size.impact.cor.schoolL	.001	.001	.001	.001	.001	.001	.001	.001	.001	.001	.001	.001
block_typeS: sd.schoolL	<.001	.001	.001	<.001	<.001	.001	.001	.001	.001	<.001	<.001	<.001
block_typeL: sd.schoolL	.001	.001	.001	<.001	.001	.001	.001	.001	.001	-.001.	.001	-.001.
block_typeS: ICC.schoolL	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001
block_typeL: ICC.schoolL	.004***	.004***	.005***	.003***	.004***	.005***	.005***	.004***	.005***	.001	.003***	.001
block_typeS: variable.nTRUE	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001
block_typeL: variable.nTRUE	<.001	<.001	.001	.003***	.004***	<.001	<.001	<.001	.001	<.001	.001	<.001
block_typeS: cv.schoolS	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001
block_typeL: cv.schoolS	.001	.001	.001	.002.	.002.	.001	.001	.001	.001	<.001	<.001	<.001
block_typeS: cv.schoolL	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001
block_typeL: cv.schoolL	.001	.001	.002**	.004***	.005***	.001	.001	.001	.002*	.001	.001	.001
block_typeS: num.cov.schoolS	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001
block_typeL: num.cov.schoolS	-.001	-.001	-.001	-.001	-.001	-.001	-.001	-.001	-.001	-.001	-.001	-.001
block_typeS: num.cov.schoolL	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001
block_typeL: num.cov.schoolL	-.001	-.001	-.001	<.001	<.001	-.001	-.001	-.001	-.001	-.001	-.001	-.001
J.barS: size.impact.cor.schoolL	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	-.001	<.001
J.barL: size.impact.cor.schoolL	-.001	-.001	-.001	-.001	-.001	-.001	-.001	-.001	-.001	-.001.	-.001*	-.001.
J.barS: sd.schoolL	<.001	<.001	-.001	-.001	-.001	-.001	-.001	<.001	-.001	-.001*	-.002**	-.001*
J.barL: sd.schoolL	-.001	-.001	-.001	-.001	-.001	-.001	-.001	-.001	-.001	-.002***	-.003***	-.002***
J.barS: ICC.schoolL	-.004***	-.004***	-.004***	-.004***	-.005***	-.005***	-.005***	-.004***	-.004***	-.005***	-.005***	-.005***
J.barL: ICC.schoolL	-.006***	-.006***	-.007***	-.008***	-.008***	-.008***	-.008***	-.006***	-.007***	-.007***	-.009***	-.007***
J.barS: variable.nTRUE	<.001	<.001	<.001	<.001	-.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001
J.barL: variable.nTRUE	-.001*	-.001*	-.002*	-.002*	-.003***	-.001	-.001.	-.001*	-.002*	-.001	-.001.	-.001
J.barS: cv.schoolS	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001
J.barL: cv.schoolS	<.001	<.001	<.001	-.001	-.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001
J.barS: cv.schoolL	-.001.	-.001.	-.002.	-.001	-.002*	-.001	-.001	-.001.	-.002.	-.001	-.001	-.001
J.barL: cv.schoolL	-.002*	-.002*	-.002**	-.002*	-.003***	-.001	-.002.	-.002*	-.002**	<.001	<.001	<.001
J.barS: num.cov.schoolS	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	.001	<.001
J.barL: num.cov.schoolS	.001	.001	.001	.001	.001	.001	.001	.001	.001	.001	.001	.001
J.barS: num.cov.schoolL	<.001	<.001	<.001	<.001	<.001	-.001	-.001	<.001	<.001	<.001	<.001	<.001
J.barL: num.cov.schoolL	<.001	<.001	-.001	-.001	-.001	-.001	-.001	<.001	-.001	<.001	<.001	<.001
size.impact.cor.schoolL: sd.schoolL	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001
size.impact.cor.schoolL: ICC.schoolL	.001*	.001*	.001.	.001*	.001*	.001.	.001.	.001*	.001.	.001.	.001.	.001.
size.impact.cor.schoolL: variable.nTRUE	.001*	.001*	.001*	.001	.001.	.001.	.001.	.001*	.001*	.001*	.001*	.001*
size.impact.cor.schoolL: cv.schoolS	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001
size.impact.cor.schoolL: cv.schoolL	.001	.001	<.001	<.001	.001	<.001	<.001	.001	<.001	<.001	<.001	<.001
size.impact.cor.schoolL: num.cov.schoolS	<.001	<.001	.001	<.001	<.001	.001	.001	<.001	.001	<.001	<.001	<.001
size.impact.cor.schoolL: num.cov.schoolL	<.001	<.001	<.001	<.001	<.001	.001	.001	<.001	.001	.001	.001	.001
sd.schoolL: ICC.schoolL	-.001**	-.001**	-.002*	-.002***	-.002**	-.002***	-.002***	-.001**	-.001*	-.003***	-.005***	-.003***

Term	LRa-FE-db	LRa-FE-het	LRa-FIcw-db	LRapw-FE-db	LRapw-FE-het	LRapw-FIpw-db	LRapw-FIpw-het	LRicw-FE-crve	LRicw-FIcw-crve	MLM-FE	MLM-FIcw	MLM-RE
sd.schoolL: variable.nTRUE	-.001*	-.001*	-.001*	-.001	<.001	<.001	<.001	-.001*	-.001*	-.001.	-.001	-.001
sd.schoolL: cv.schoolS	-.002**	-.002**	-.002*	-.002*	-.002*	-.002*	-.001*	-.002**	-.002*	-.001*	-.001.	-.001.
sd.schoolL: cv.schoolL	-.001*	-.001*	-.001*	<.001	<.001	<.001	<.001	-.001*	-.001*	-.001	<.001	<.001
sd.schoolL: num.cov.schoolS	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001
sd.schoolL: num.cov.schoolL	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001
ICC.schoolL: variable.nTRUE	-.001*	-.001*	-.001*	.003***	.003***	.002***	.003***	-.002**	-.002**	<.001	<.001	<.001
ICC.schoolL: cv.schoolS	<.001	<.001	<.001	.001	.001	.001	.001	<.001	<.001	<.001	<.001	<.001
ICC.schoolL: cv.schoolL	-.003***	-.003***	-.003***	.003***	.003***	.002*	.002**	-.003***	-.003***	<.001	-.001	-.001
ICC.schoolL: num.cov.schoolS	-.001	-.001	-.001	-.001	-.001	-.001	-.001	-.001	-.001	-.001	-.001	-.001
ICC.schoolL: num.cov.schoolL	<.001	<.001	-.001	<.001	<.001	-.001	-.001	<.001	-.001	<.001	<.001	<.001
variable.nTRUE: cv.schoolS	<.001	<.001	<.001	.002*	.002**	<.001	.001	<.001	<.001	<.001	<.001	<.001
variable.nTRUE: cv.schoolL	.004***	.004***	.005***	.006***	.008***	.003***	.005***	.005***	.006***	.001	.001*	.001
variable.nTRUE: num.cov.schoolS	-.001*	-.001*	-.002*	-.001*	-.002*	-.002*	-.002*	-.001*	-.002*	-.001*	-.001.	-.001*
variable.nTRUE: num.cov.schoolL	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001
cv.schoolS: num.cov.schoolS	.001.	.001.	.002*	.001	.001	.002*	.002*	.001.	.002*	.001	.001	.001
cv.schoolL: num.cov.schoolS	.002*	.002*	.002*	.002*	.002*	.002*	.002*	.002*	.002*	.001*	.001*	.001*
cv.schoolS: num.cov.schoolL	<.001	<.001	.001	<.001	<.001	.001	.001	<.001	.001	<.001	<.001	<.001
cv.schoolL: num.cov.schoolL	.001	.001	.001	.001	.001	.001	.001	.001	.001	.001	.001	.001

*Significance: '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1*

Top Five Scenarios with Most SE Estimate Instability by Method

method	sd.SE hat.s	J.bar	sd school	ICC school	cv school	num.cov school	size.impact cor.school	block type
LRapw-FE-db	0.0585	0	L	L	L	0	L	L
LRapw-FE-db	0.0585	0	L	L	L	L	L	L
LRapw-FE-db	0.0582	0	0	L	L	L	L	L
LRapw-FE-db	0.0573	0	0	L	L	0	L	L
LRapw-FE-db	0.0569	0	0	L	S	L	L	L
LRapw-FE-het	0.0717	0	L	L	L	0	L	L
LRapw-FE-het	0.0715	0	L	L	L	L	L	L
LRapw-FE-het	0.0715	0	0	L	L	L	L	L
LRapw-FE-het	0.0673	0	0	L	L	0	L	L
LRapw-FE-het	0.0638	0	L	L	L	0	0	L
MLM-FIcw	0.0519	0	L	L	0	L	L	L
MLM-FIcw	0.0516	0	L	L	0	0	L	L

method	sd.SE hat.s	J.bar	sd school	ICC school	cv school	num.cov school	size.impact cor.school	block type
LRa-FIcw-db	0.0659	S	0	L	S	S	0	L
LRa-FIcw-db	0.0612	0	L	L	L	L	L	L
LRa-FIcw-db	0.056	0	L	L	S	S	0	L
LRa-FIcw-db	0.0541	0	L	L	L	0	L	L
LRa-FIcw-db	0.0519	0	L	L	S	0	0	L
LRapw-FIpw-db	0.0659	S	0	L	S	S	0	L
LRapw-FIpw-db	0.0565	0	L	L	L	L	L	L
LRapw-FIpw-db	0.056	0	L	L	S	S	0	L
LRapw-FIpw-db	0.0519	0	L	L	S	0	0	L
LRapw-FIpw-db	0.0519	0	L	L	0	L	L	L
LRapw-FIpw-het	0.0659	S	0	L	S	S	0	L
LRapw-FIpw-het	0.0579	0	L	L	L	L	L	L
LRapw-FIpw-het	0.056	0	L	L	S	S	0	L
LRapw-FIpw-het	0.0521	0	L	L	L	0	L	L
LRapw-FIpw-het	0.0519	0	L	L	S	0	0	L
LRicw-FIcw-crve	0.0659	S	0	L	S	S	0	L
LRicw-FIcw-crve	0.0613	0	L	L	L	L	L	L
LRicw-FIcw-crve	0.056	0	L	L	S	S	0	L
LRicw-FIcw-crve	0.0543	0	L	L	L	0	L	L
LRicw-FIcw-crve	0.0519	0	L	L	S	0	0	L

Top Five Scenarios with Most Normalized SE Estimate Instability by Method

method	sd.SE hat.s.norm	sd.SE hat.s	J.bar	sd school	ICC school	cv school	num.cov school	size.impact cor.school	block type
LRa-FE-db	2.1	0.01	L	0	0	L	S	0	0
LRa-FE-db	2.06	0.00985	L	0	0	L	L	0	0

method	sd.SE hat.s.norm	sd.SE hat.s	J.bar	sd school	ICC school	cv school	num.cov school	size.impact cor.school	block type
LRa-FE-db	2.04	0.00986	L	0	0	L	L	L	0
LRa-FE-db	2.03	0.00951	L	0	0	L	L	L	S
LRa-FE-db	2.03	0.00791	L	0	0	L	L	0	S
LRa-FE-het	2.1	0.01	L	0	0	L	S	0	0
LRa-FE-het	2.06	0.00985	L	0	0	L	L	0	0
LRa-FE-het	2.04	0.00986	L	0	0	L	L	L	0
LRa-FE-het	2.03	0.00792	L	0	0	L	L	0	S
LRa-FE-het	2.03	0.00951	L	0	0	L	L	L	S
LRapw-FE-db	1.49	0.0198	0	L	0	S	S	0	L
LRapw-FE-db	1.47	0.0155	S	0	0	S	L	0	L
LRapw-FE-db	1.44	0.0122	L	0	L	L	L	0	L
LRapw-FE-db	1.42	0.0247	S	0	0	S	0	L	L
LRapw-FE-db	1.41	0.0197	L	0	L	L	0	L	L
LRapw-FE-het	1.52	0.0337	0	0	L	L	S	0	L
LRapw-FE-het	1.51	0.0361	L	0	L	L	S	0	L
LRapw-FE-het	1.49	0.0127	L	0	L	L	L	0	L
LRapw-FE-het	1.48	0.0589	0	0	L	L	L	0	L
LRapw-FE-het	1.48	0.0206	L	0	L	L	0	L	L
LRicw-FE-crve	2.16	0.0103	L	0	0	L	S	0	0
LRicw-FE-crve	2.12	0.0102	L	0	0	L	L	0	0
LRicw-FE-crve	2.1	0.0102	L	0	0	L	L	L	0
LRicw-FE-crve	2.09	0.0098	L	0	0	L	L	L	S
LRicw-FE-crve	2.08	0.00812	L	0	0	L	L	0	S
MLM-FIcw	1.15	0.0148	0	L	0	0	0	0	0
MLM-FIcw	1.15	0.0146	0	L	0	0	0	L	0
MLM-FIcw	1.15	0.0148	0	L	0	S	0	L	0

method	sd.SE hat.s.norm	sd.SE hat.s	J.bar	sd school	ICC school	cv school	num.cov school	size.impact cor.school	block type
MLM-FIcw	1.15	0.0148	0	L	0	0	S	0	0
MLM-FIcw	1.15	0.0147	0	L	0	0	L	L	0
LRa-FIcw-db	2.12	0.0101	L	0	0	L	S	0	0
LRa-FIcw-db	2.07	0.00991	L	0	0	L	L	0	0
LRa-FIcw-db	2.06	0.00995	L	0	0	L	L	L	0
LRa-FIcw-db	2.05	0.00801	L	0	0	L	L	0	S
LRa-FIcw-db	2.05	0.00962	L	0	0	L	L	L	S
LRapw-FIpw-db	1.38	0.0388	S	L	0	0	0	L	L
LRapw-FIpw-db	1.34	0.0659	S	0	L	S	S	0	L
LRapw-FIpw-db	1.33	0.0369	0	0	L	0	S	L	L
LRapw-FIpw-db	1.33	0.0143	L	L	L	L	L	0	0
LRapw-FIpw-db	1.33	0.0268	L	0	L	L	S	L	L
LRapw-FIpw-het	1.42	0.0152	L	L	L	L	L	0	0
LRapw-FIpw-het	1.41	0.0154	L	L	L	L	0	0	0
LRapw-FIpw-het	1.39	0.0152	L	L	L	L	S	0	0
LRapw-FIpw-het	1.39	0.0177	L	L	L	L	L	0	S
LRapw-FIpw-het	1.38	0.014	L	0	L	L	0	L	0
LRicw-FIcw-crve	2.21	0.0106	L	0	0	L	S	0	0
LRicw-FIcw-crve	2.17	0.0104	L	0	0	L	L	0	0
LRicw-FIcw-crve	2.15	0.0104	L	0	0	L	L	L	0
LRicw-FIcw-crve	2.14	0.01	L	0	0	L	L	L	S
LRicw-FIcw-crve	2.13	0.00831	L	0	0	L	L	0	S

Are the estimated standard errors targeting the finite or super population

*This section answers the questions on page 39. I can answer: 1) how accurate are the estimated standard errors, 2) how stable are the standard errors and 3) are certain standard errors performing better than others. I'm not

- The Choosing an estimand and excluding the fully interacted estimators results in estimated standards errors that are less variable across estimators. However, it is notable that differences are still often non-trivial. Even in Figure we see estimated standard errors are 20 percent different across estimators in 24 percent of the cases we examined.

[ADD ANY TEXT WE CAN HERE ABOUT WHAT WE'VE LEAREND FROM SIMULATIONS. IS THIS BECAUSE TRUE SEs VARY ACROSS ESTIMATORS? IS THIS ESTIMATION ERROR? ARE CERTAIN SE ESTIMATORS MORE ACCURATE THAN OTHERS?]

*make sure I answer this question when I finish my write up of the standard errors

*pg 39

*The estimated standard errors for the unbiased cluster targeted estimators are consistently good and are also pretty good for the unbiased person targeted estimators, although in some of the scenarios they're under estimated. The biased estimators are often also overestimating the standard errors.

*Graphs in this section back up the point on pg. X of the paper

*True super population standard errors are almost always larger than the finite population standard errors, the cases where finite standard errors are larger it seems almost certainly to be noise in the simulations.

*Approximately a quarter of the time the true finite and true super standard errors are the same.

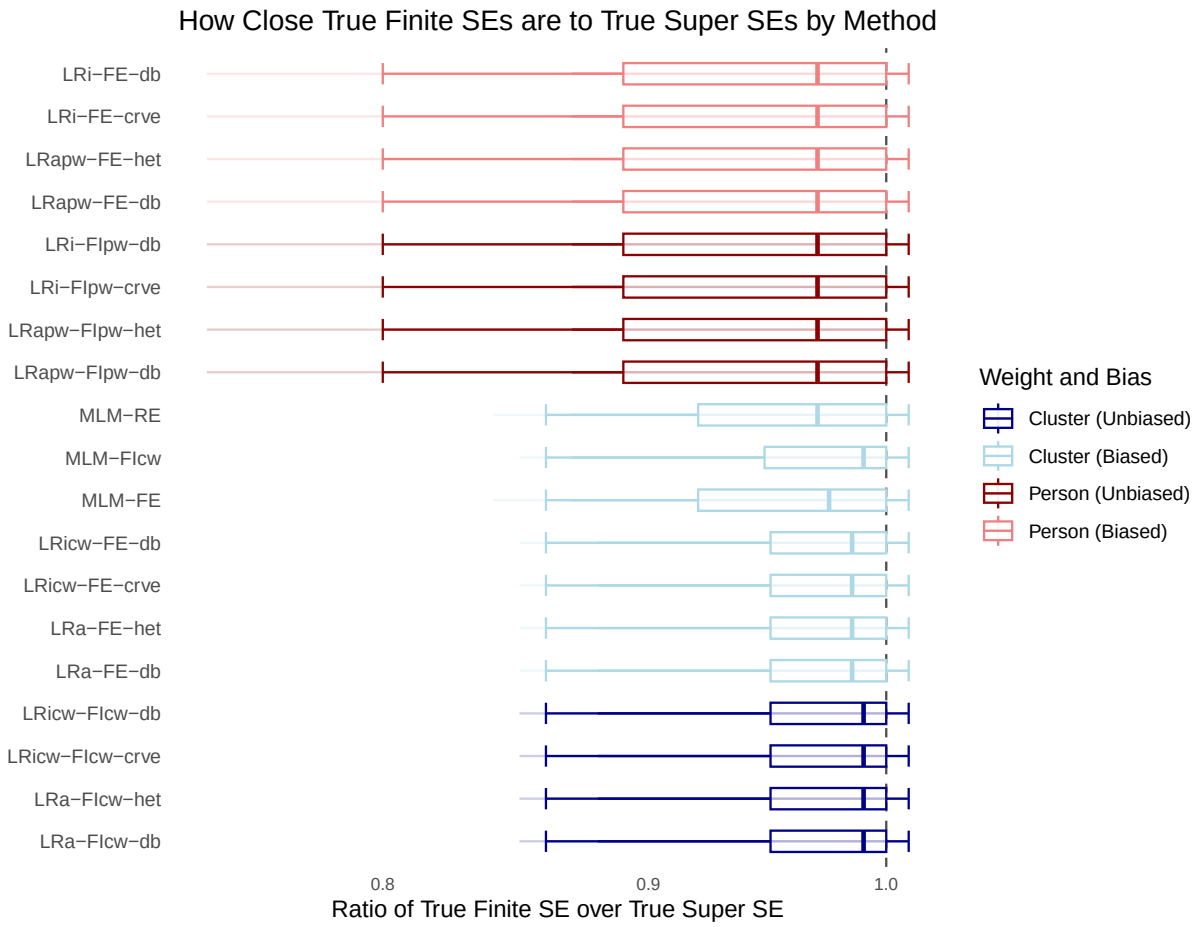
*Approximately three quarters of the time the estimated standard errors are closer to the super population standard errors than the finite population standard errors,

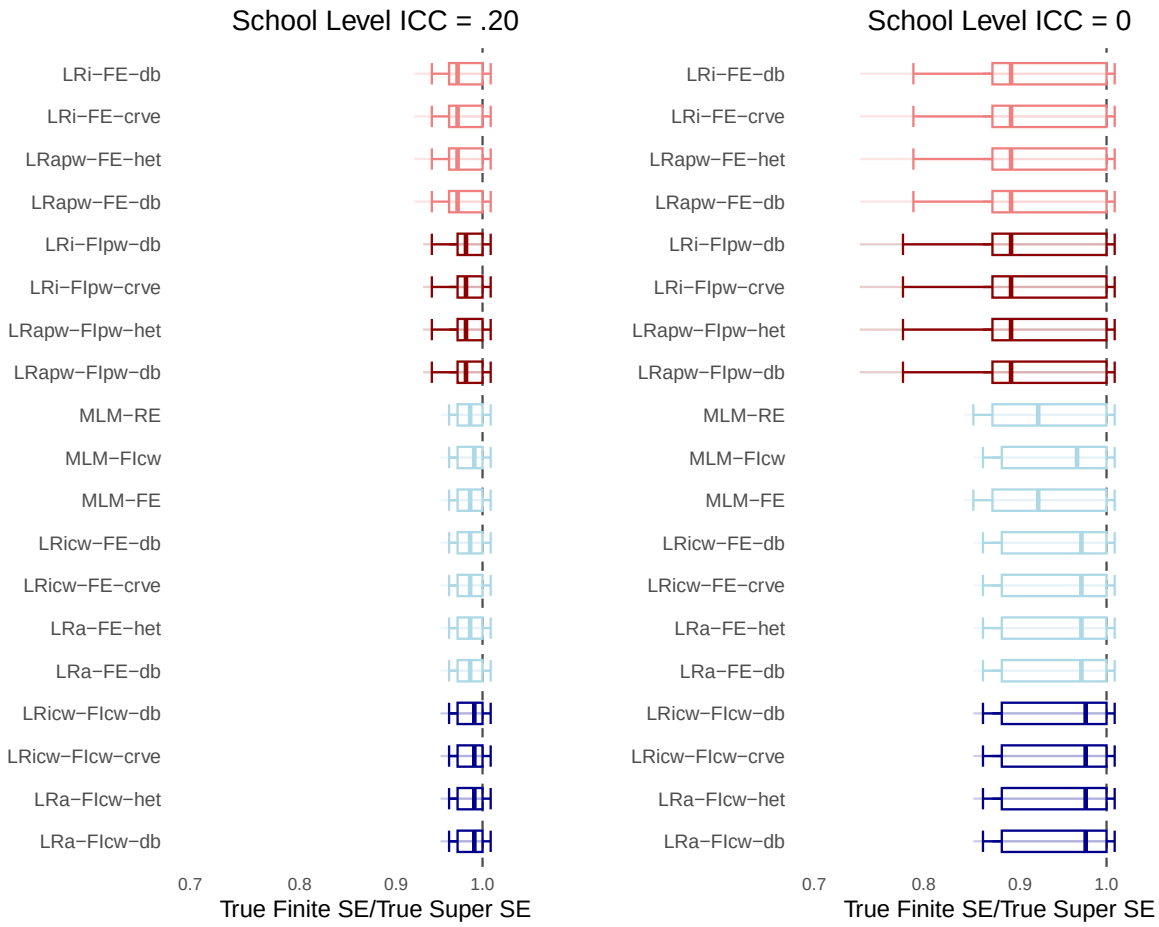
-

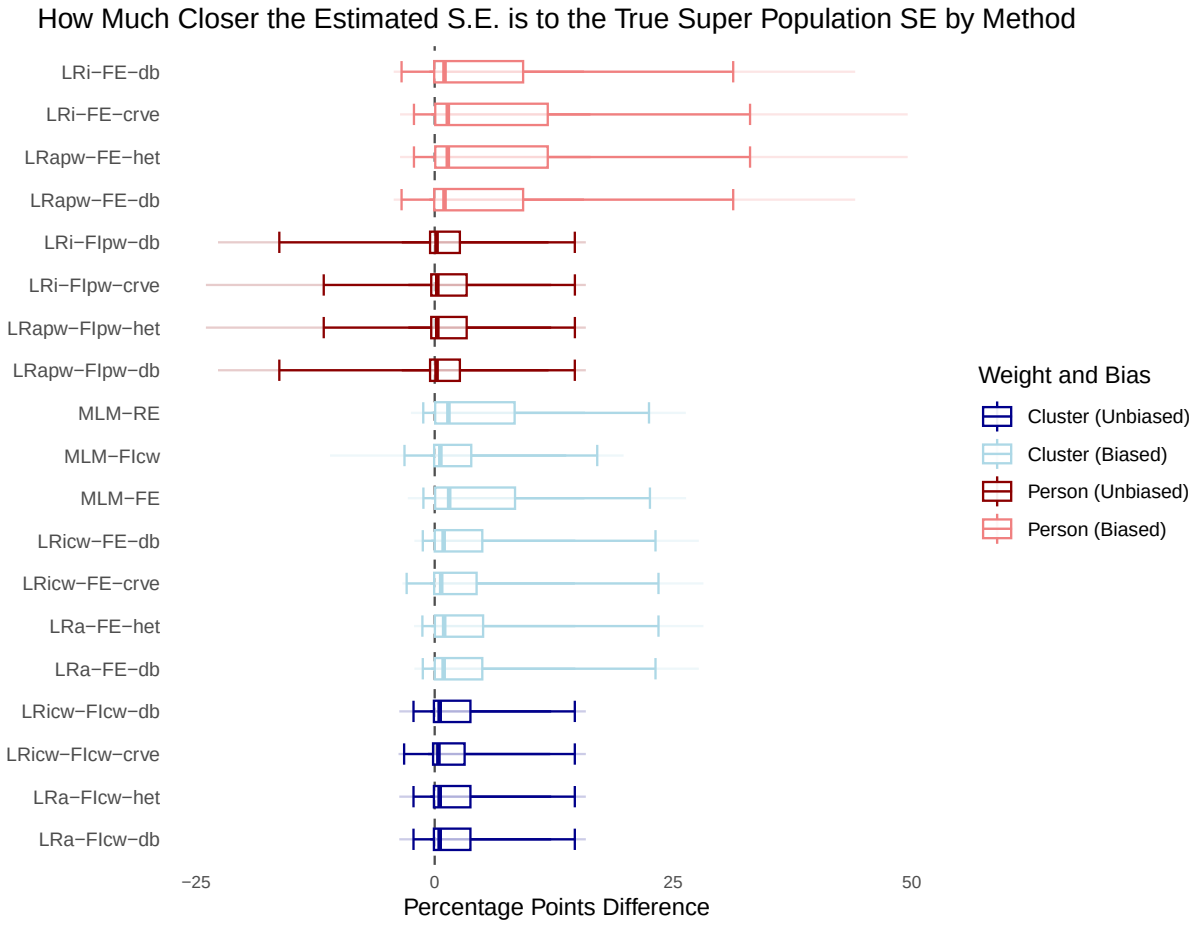
Standard error inflation is the ratio of the estimated standard error over the true standard error. High inflation means the standard errors are too big and low inflation means the standard errors are too small.

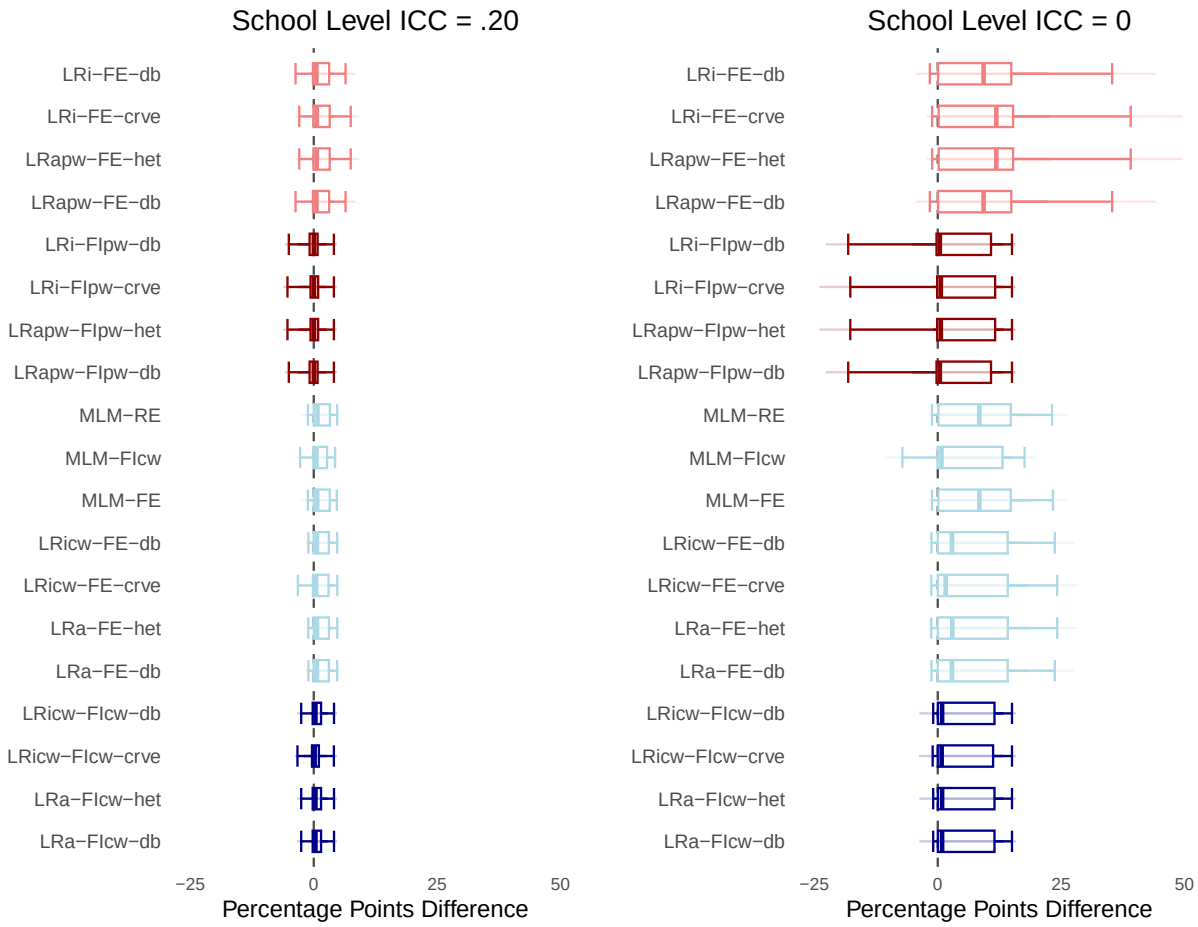
For some of the methods the standard error estimates are way off, but it does not seem obviously driven by low degrees of freedom. The true finite standard errors do not look to be the same as the true super standard errors.

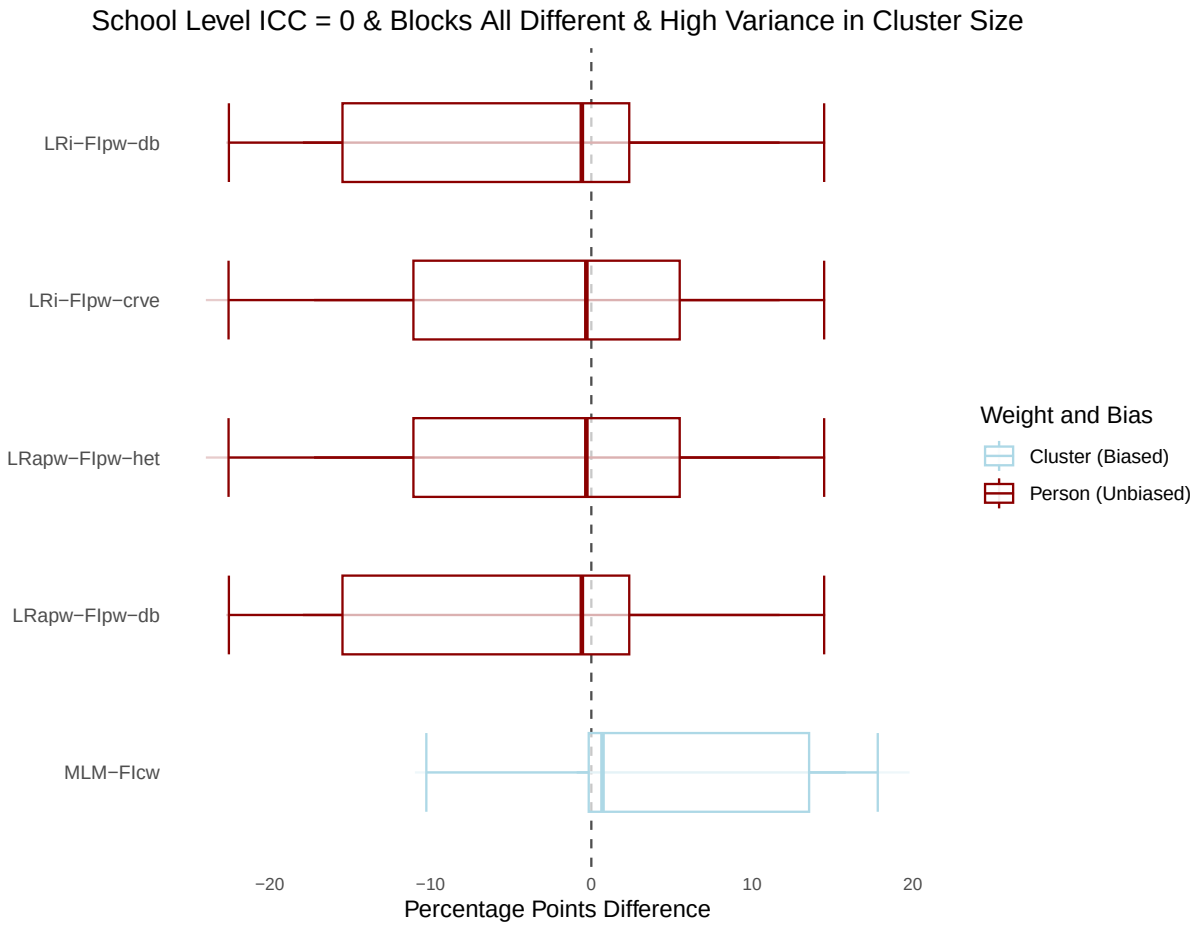
#To Do *make edits to all the graphs* move the finite versus super population stuff to its own section *fix the formating of the table* add in description of the plots


How Close True Finite SEs are to True Super SEs by Method
LRi−FE−db
LRi−FE−crve
LRapw−FE−het
LRapw−FE−db
LRi−FIpw−db
LRi−FIpw−crve
LRapw−FIpw−het
LRapw−FIpw−db
MLM−RE
MLM−FIcw
MLM−FE
LRicw−FE−db
LRicw−FE−crve
LRa−FE−het
LRa−FE−db
LRicw−FIcw−db
LRicw−FIcw−crve
LRa−FIcw−het
LRa−FIcw−db
0.8
0.9
1.0
Ratio of True Finite SE over True Super SE
Weight and Bias
Cluster (Unbiased)
Cluster (Biased)
Person (Unbiased)
Person (Biased)


School Level ICC = .20
School Level ICC = 0
LRi−FE−db
LRi−FE−crve
LRapw−FE−het
LRapw−FE−db
LRi−FIpw−db
LRi−FIpw−crve
LRapw−FIpw−het
LRapw−FIpw−db
MLM−RE
MLM−FIcw
MLM−FE
LRicw−FE−db
LRicw−FE−crve
LRa−FE−het
LRa−FE−db
LRicw−FIcw−db
LRicw−FIcw−crve
LRa−FIcw−het
LRa−FIcw−db
0.7
0.8
0.9
1.0
True Finite SE/True Super SE


How Much Closer the Estimated S.E. is to the True Super Population SE by Method
LRi−FE−db
LRi−FE−crve
LRapw−FE−het
LRapw−FE−db
LRi−FIpw−db
LRi−FIpw−crve
LRapw−FIpw−het
LRapw−FIpw−db
MLM−RE
MLM−FIcw
MLM−FE
LRicw−FE−db
LRicw−FE−crve
LRa−FE−het
LRa−FE−db
LRicw−FIcw−db
LRicw−FIcw−crve
LRa−FIcw−het
LRa−FIcw−db
−25
0
25
50
Percentage Points Difference
Weight and Bias
Cluster (Unbiased)
Cluster (Biased)
Person (Unbiased)
Person (Biased)


School Level ICC = .20
School Level ICC = 0
LRi−FE−db
LRi−FE−crve
LRapw−FE−het
LRapw−FE−db
LRi−FIpw−db
LRi−FIpw−crve
LRapw−FIpw−het
LRapw−FIpw−db
MLM−RE
MLM−FIcw
MLM−FE
LRicw−FE−db
LRicw−FE−crve
LRa−FE−het
LRa−FE−db
LRicw−FIcw−db
LRicw−FIcw−crve
LRa−FIcw−het
LRa−FIcw−db
−25
0
25
50
Percentage Points Difference



Top Five Scenarios with Where Estimated SEs are Closest to Targeting Finite SEs

method	dif	infla te.f	infla te.s	true se.ratio	J.bar	sd school	ICC school	cv school	num.cov school	size.impact cor.school	block type
MLM-FIcw	-11	0.986	0.877	0.89	L	L	0	L	L	0	L
MLM-FIcw	-9.98	0.978	0.878	0.9	S	L	0	L	L	L	L
MLM-FIcw	-9.74	0.963	0.866	0.9	S	L	0	L	0	0	L
MLM-FIcw	-9.57	0.973	0.877	0.9	0	L	0	S	S	0	L
MLM-FIcw	-8.31	0.928	0.844	0.91	S	L	0	0	0	L	L
LRapw-FIpw-db	-22.7	0.943	0.716	0.76	S	0	0	L	L	L	L
LRapw-FIpw-db	-22.5	1.01	0.766	0.76	0	L	0	L	L	L	L
LRapw-FIpw-db	-22.2	0.987	0.765	0.78	S	L	0	L	S	0	L
LRapw-FIpw-db	-21.4	1.02	0.763	0.75	0	L	0	L	0	0	L
LRapw-FIpw-db	-20.9	0.933	0.723	0.78	L	0	0	L	S	0	L
LRapw-FIpw-het	-24	0.996	0.756	0.76	S	0	0	L	L	L	L
LRapw-FIpw-het	-22	0.981	0.761	0.78	L	0	0	L	S	0	L

method	dif	infla te.f	infla te.s	true se.ratio	J.bar	sd school	ICC school	cv school	num.cov school	size.impact cor.school	block type
LRapw-FIpw-het	-21	0.977	0.767	0.79	0	0	0	L	S	0	L
LRapw-FIpw-het	-19.1	0.976	0.785	0.8	0	0	0	L	0	0	L
LRapw-FIpw-het	-18.7	0.968	0.781	0.81	0	0	0	L	S	L	L

What do the confidence intervals look like for each estimator

Main Bullet Points:

*The unbiased estimators also have less error than the biased estimators. The unbiased person estimator has less instability than the person estimator, but the unbiased cluster estimator has more instability.

*The MLM estimators are the most stable, which fits with theory.

*High RMSE and estimate instability is driven by a low number of clusters, the interaction of a low number of clusters and high cluster level ICC, and high cluster level treatment effect variance.

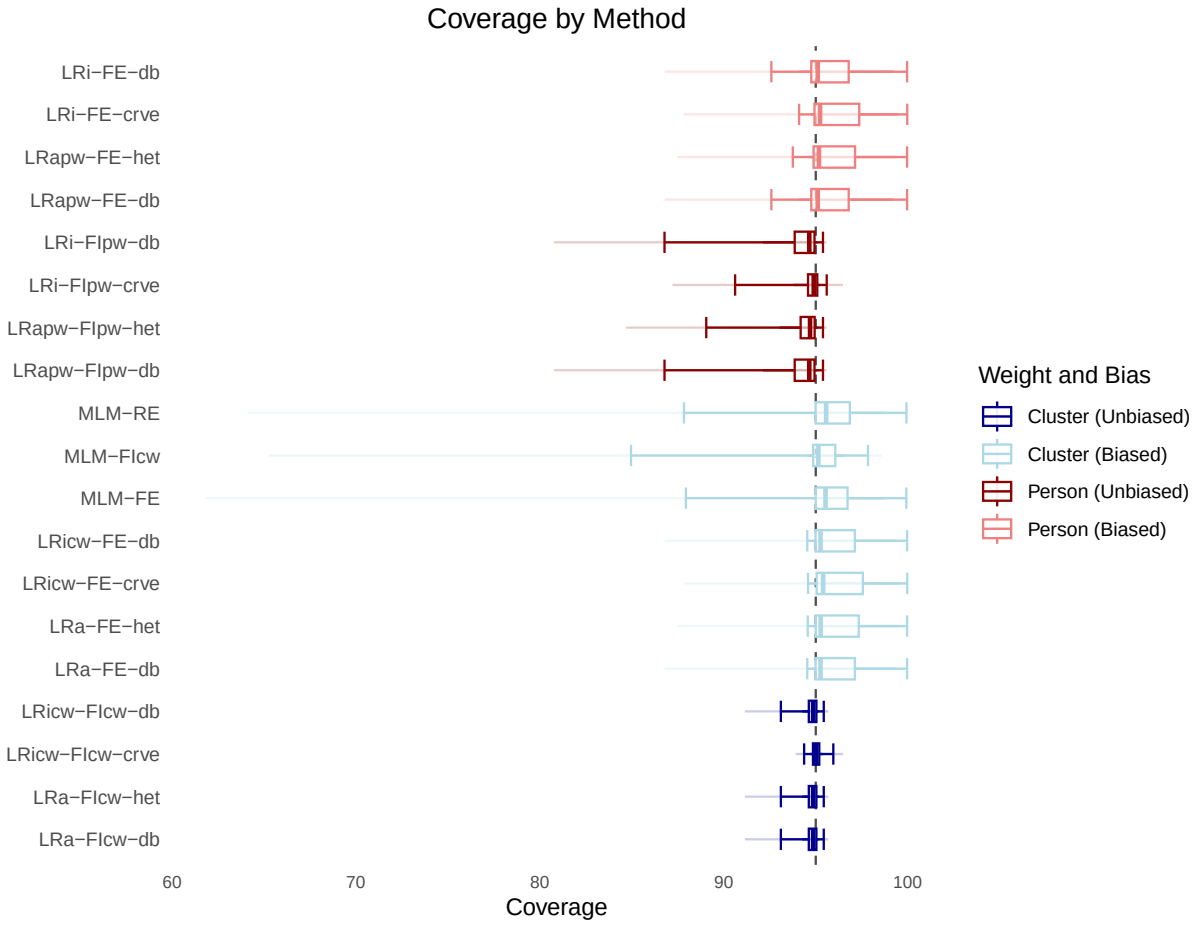
Description of Analysis:

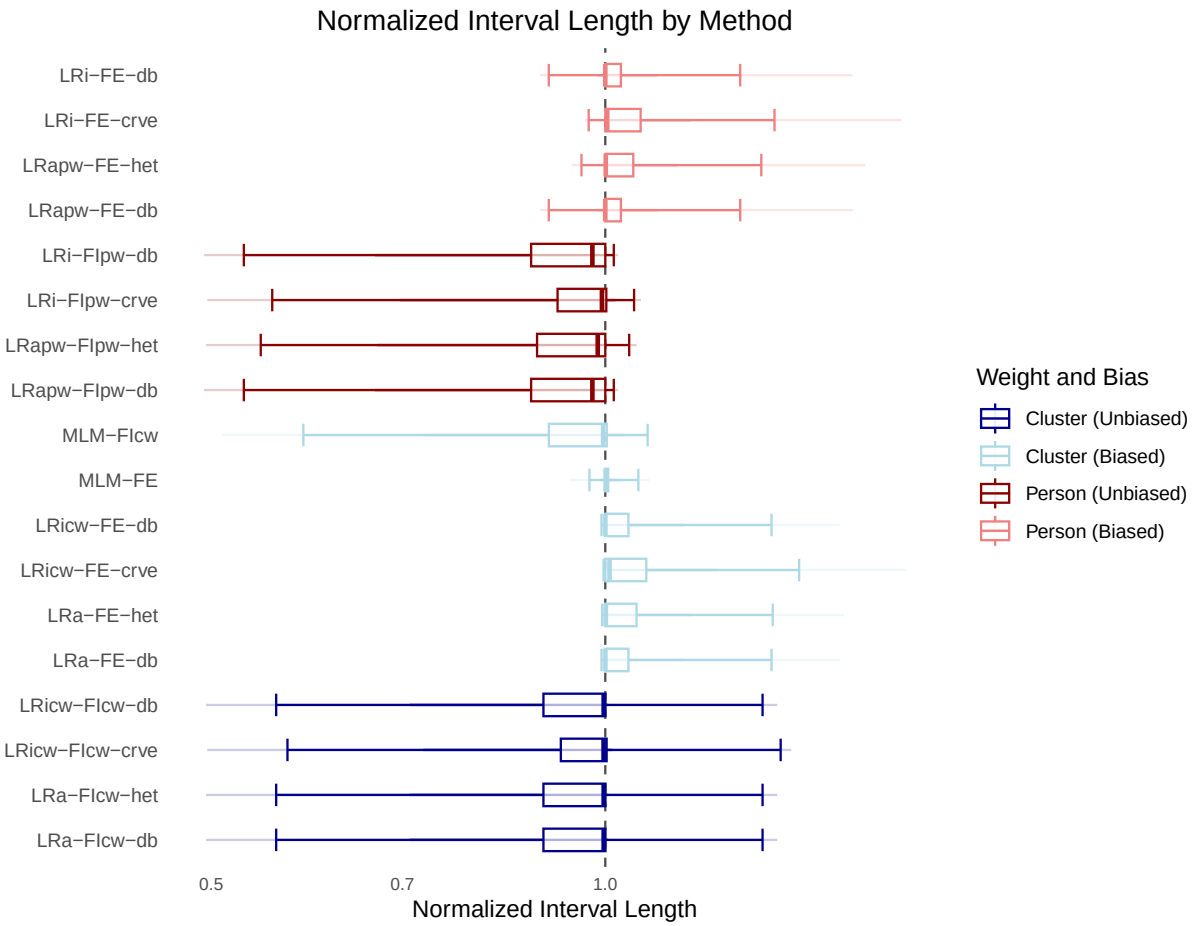
*The first graph is a boxplot of the normalized RMSE across the different scenarios by method. The normalized RMSE is the RMSE for given method divided by the median RMSE across methods within a given simulation scenario. Therefore each data point is adjusted for the underlying base RMSE from a given set of DGPs and represents the RMSE driven by the estimation method.

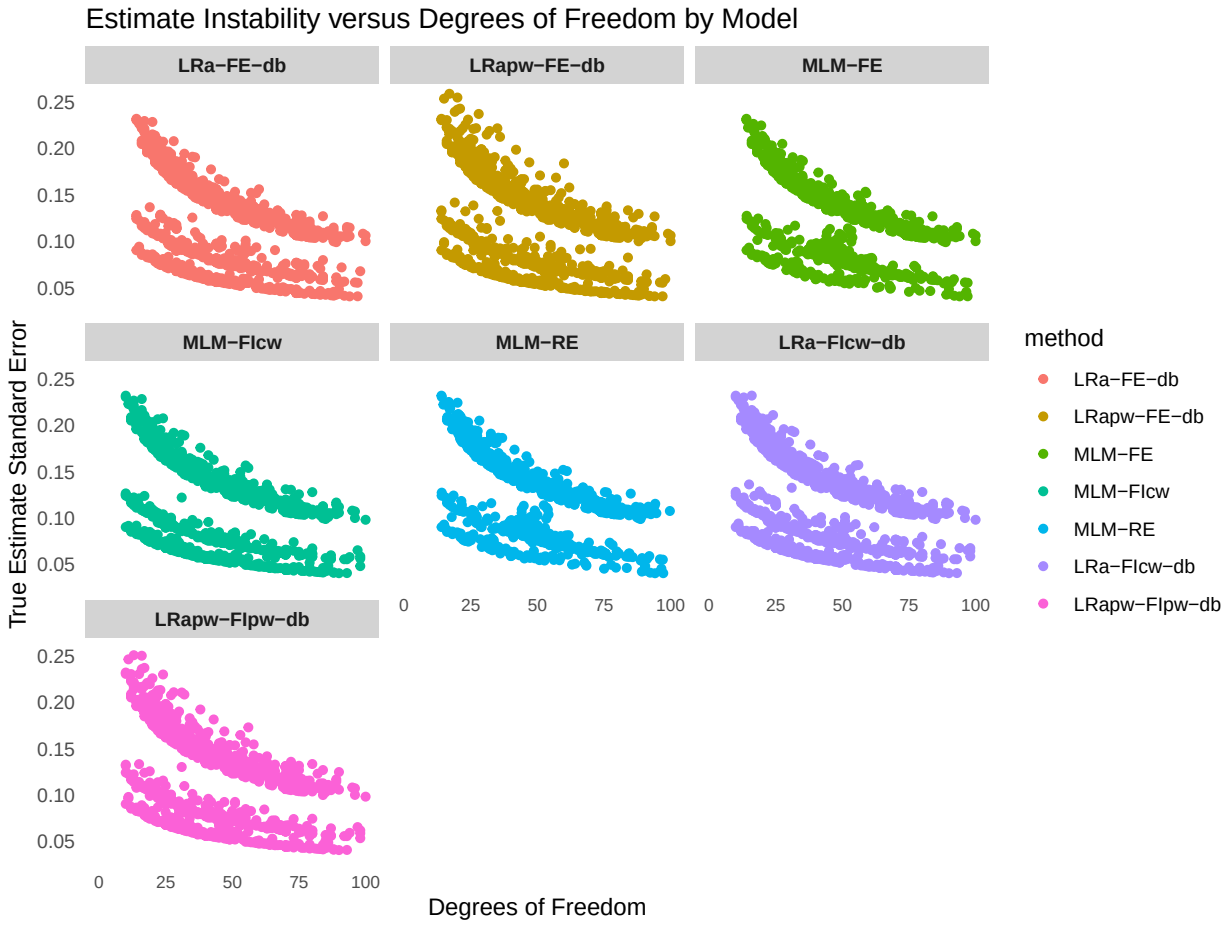
*The second graph is a boxplot of the normalized estimate standard deviation across the different scenarios by method. This is also normalized by dividing by the median estimate standard deviation within a given simulation scenario.

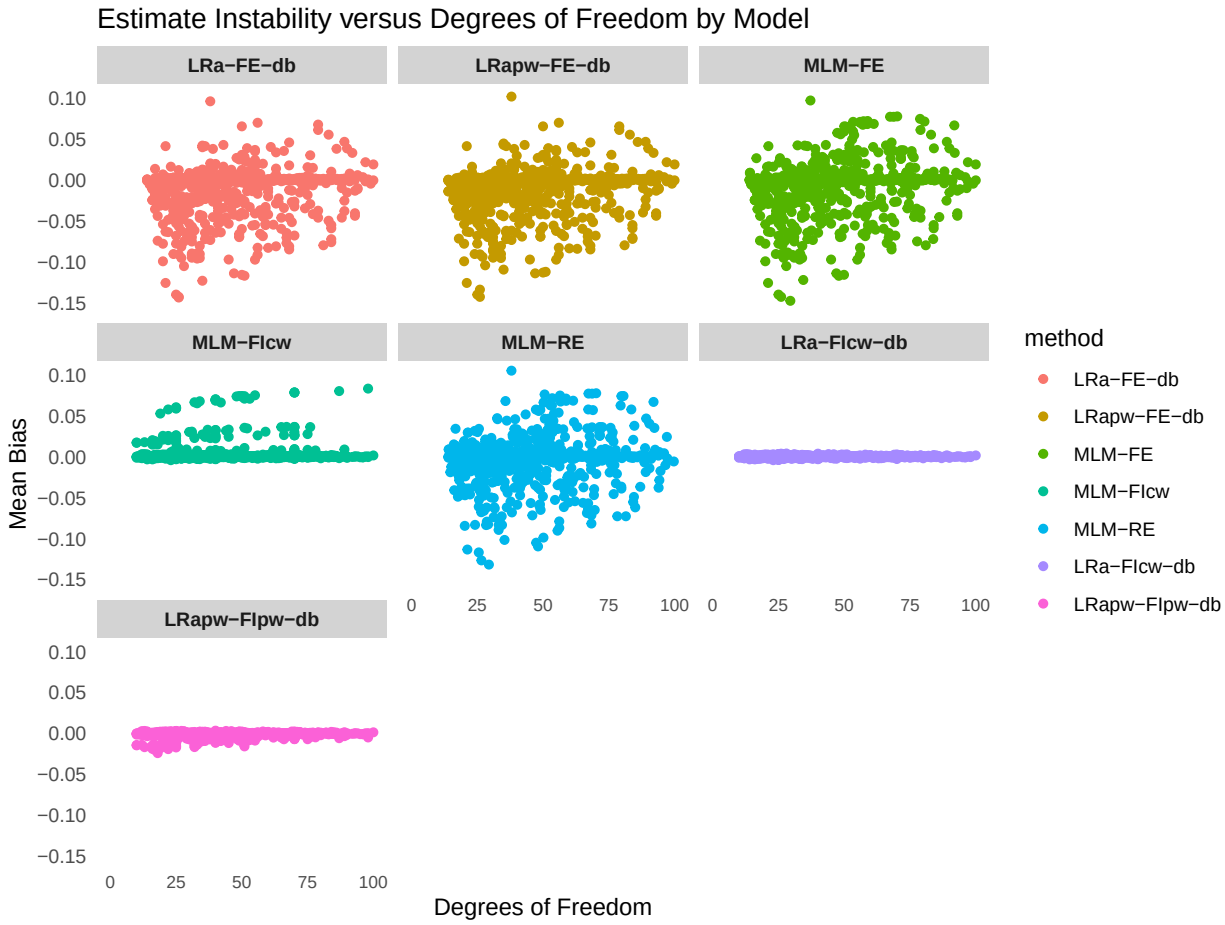
*The first table contains the regression coefficients from a regression that predicts the RMSE using different data generating parameters and the second table is from regressions predicting the estimate standard deviation. Each regression is run separately by estimator and the RMSE is against the targeted estimator.

Coverage is the percentage of simulation replications where the estimator is within the 95% confidence interval. We would like the number to 95% (the boxplots have a red line at there). The notable things are that the coverage for the MLM estimators is often very bad but other than that the coverage for the cluster weighted estimators is generally good. The coverage rates for the person weighted estimators however are pretty bad across methods,it's either very low (meaning the SEs are being underestimated) or too high (meaning the SEs are being overestimated within the same methods. In addition, finite versus super makes very little difference for the cluster targeted estimators but makes a noticeable difference for the person weighted estimators.


Coverage by Method
LRi−FE−db
LRi−FE−crve
LRapw−FE−het
LRapw−FE−db
LRi−FIpw−db
LRi−FIpw−crve
LRapw−FIpw−het
LRapw−FIpw−db
MLM−RE
MLM−FIcw
MLM−FE
LRicw−FE−db
LRicw−FE−crve
LRa−FE−het
LRa−FE−db
LRicw−FIcw−db
LRicw−FIcw−crve
LRa−FIcw−het
LRa−FIcw−db
60
70
80
90
100
Coverage
Weight and Bias
Cluster (Unbiased)
Cluster (Biased)
Person (Unbiased)
Person (Biased)


Normalized Interval Length by Method
LRi−FE−db
LRi−FE−crve
LRapw−FE−het
LRapw−FE−db
LRi−FIpw−db
LRi−FIpw−crve
LRapw−FIpw−het
LRapw−FIpw−db
MLM−FIcw
MLM−FE
LRicw−FE−db
LRicw−FE−crve
LRa−FE−het
LRa−FE−db
LRicw−FIcw−db
LRicw−FIcw−crve
LRa−FIcw−het
LRa−FIcw−db
0.5
0.7
1.0
Normalized Interval Length
Weight and Bias
Cluster (Unbiased)
Cluster (Biased)
Person (Unbiased)
Person (Biased)


Estimate Instability versus Degrees of Freedom by Model
LRa−FE−db
LRapw−FE−db
MLM−FE
MLM−FIcw
MLM−RE
LRa−FIcw−db
LRapw−FIpw−db
True Estimate Standard Error
0.25
0.20
0.15
0.10
0.05
0
25
50
75
100
Degrees of Freedom
method
LRa−FE−db
LRapw−FE−db
MLM−FE
MLM−FIcw
MLM−RE
LRa−FIcw−db
LRapw−FIpw−db


Estimate Instability versus Degrees of Freedom by Model
LRa−FE−db
LRapw−FE−db
MLM−FE
MLM−FIcw
MLM−RE
LRa−FIcw−db
LRapw−FIpw−db
Mean Bias
0.10
0.05
0.00
−0.05
−0.10
−0.15
0
25
50
75
100
Degrees of Freedom
method
LRa−FE−db
LRapw−FE−db
MLM−FE
MLM−FIcw
MLM−RE
LRa−FIcw−db
LRapw−FIpw−db